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(54) **BUCKWHEAT-DERIVED
C-GLYCOSYLTRANSFERASE GENE AND
UTILIZATION THEREOF**

FOREIGN PATENT DOCUMENTS

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OTHER PUBLICATIONS

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GenBank: AB909376, 2014, <https://www.ncbi.nlm.nih.gov/nuccore/7AB909376>.*

GenBank: AB909375, 2014, <https://www.ncbi.nlm.nih.gov/nuccore/AB909375.1>.*

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GenBank Accession: AB909375.1 <https://www.ncbi.nlm.nih.gov/nuccore/AB909375>; Available: Nov. 18, 2014 (Year: 2014).*

GenBank Accession: X77943; <https://www.ncbi.nlm.nih.gov/nuccore/X77943>; Available Nov. 14, 2006 (Year: 2006).*

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 69 days.

Chen et al., ACS Catalysis 8.6 (2018): 4917-4927. (Year: 2018).*

Poulos et al., Cytochrome P450: structure, mechanism, and biochemistry (2015): 3-32. (Year: 2015).*

Nagatomo et al., The Plant Journal 80.3 (2014): 437-448. (Year: 2014).*

(21) Appl. No.: **17/765,692**

Yabuya, et al., Euphytica 98 (1997): 163-167. (Year: 1997).*

(22) PCT Filed: **Sep. 25, 2020**

Jiang et al., Plants 5.2 (2016): 27. (Year: 2016).*

(86) PCT No.: **PCT/JP2020/036451**

Tanaka et al., Fifty years of cytochrome P450 research (2014): 207-229. (Year: 2014).*

§ 371 (c)(1),

Sugio et al., Journal of bioscience and bioengineering 105.3 (2008): 300-302. (Year: 2008).*

(2) Date: **Mar. 31, 2022**

Chandler et al., Biotechnology letters 33.2 (2011): 207-214. (Year: 2011).*

(87) PCT Pub. No.: **WO2021/065749**

Liu et al., Plant Cell 32.9 (2020): 2917-2931. (Year: 2020).*

PCT Pub. Date: **Apr. 8, 2021**

Guo et al., 2004, Proc. Natl. Acad. Sci. USA 101: 9205-9210. (Year: 2004).*

(65) **Prior Publication Data**

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Katsumoto et al., Plant and cell physiology 48.11 (2007): 1589-1600 (Year: 2007).*

(30) **Foreign Application Priority Data**

Oct. 1, 2019 (JP) 2019-181693

Akashi et al., Plant Physiology 137.3 (2005): 882-891 (Year: 2005).*

(51) **Int. Cl.**

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A01H 6/74 (2018.01)

C12N 9/02 (2006.01)

C12N 9/10 (2006.01)

C12N 9/88 (2006.01)

C12N 15/82 (2006.01)

Rauter et al., "C-Glycosylflavonoids: Identification, Bioactivity and Synthesis", Natural Product Communications, Jan. 1, 2007, vol. 2, No. 11, pp. 1175-1196, cited in EP Search Report dated Jan. 3, 2024. (22 pages).

Nagatomo, Y. et al., "Purification, molecular cloning and functional characterization of flavonoid C-glucosyltransferases from *Fagopyrum esculentum* M. (buckwheat) cotyledon", The Plant Journal, 2014, vol. 80, pp. 437-448; Cited in ISR. (12 pages).

International Search Report dated Nov. 24, 2020, issued in counterpart Application No. PCT/JP2020/036451. (3 pages).

Arabidopsis thaliana ortholog of sugar beet HS1 PRO-12 (HSPRO2), mRNA, NCBI Reference Sequence: NM 129558.3. (3 pages); cited in CN Office Action dated Jul. 8, 2024.

(52) **U.S. Cl.**

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(2013.01); **A01H 6/749** (2018.05); **C12N**

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(2013.01); **C12Y 11/4/14** (2013.01); **C12Y**

201/01 (2013.01); **C12Y 204/01** (2013.01);

C12Y 402/01105 (2013.01)

* cited by examiner

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(58) **Field of Classification Search**

None

See application file for complete search history.

(57) **ABSTRACT**

Transgenic plants with blue flower color, or their inbred or outbred progeny, or their propagules, partial plant bodies, tissues or cells, are provided. A buckwheat-derived C-glucosyltransferase (CGT) gene or its homolog is transferred into a host plant to cause delphinidin-type anthocyanins and flavone mono-C-glycosides to be copresent in the plant cells.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2020/0283782 A1 9/2020 Nakamura et al.

11 Claims, 6 Drawing Sheets

Specification includes a Sequence Listing.

FIG. 1

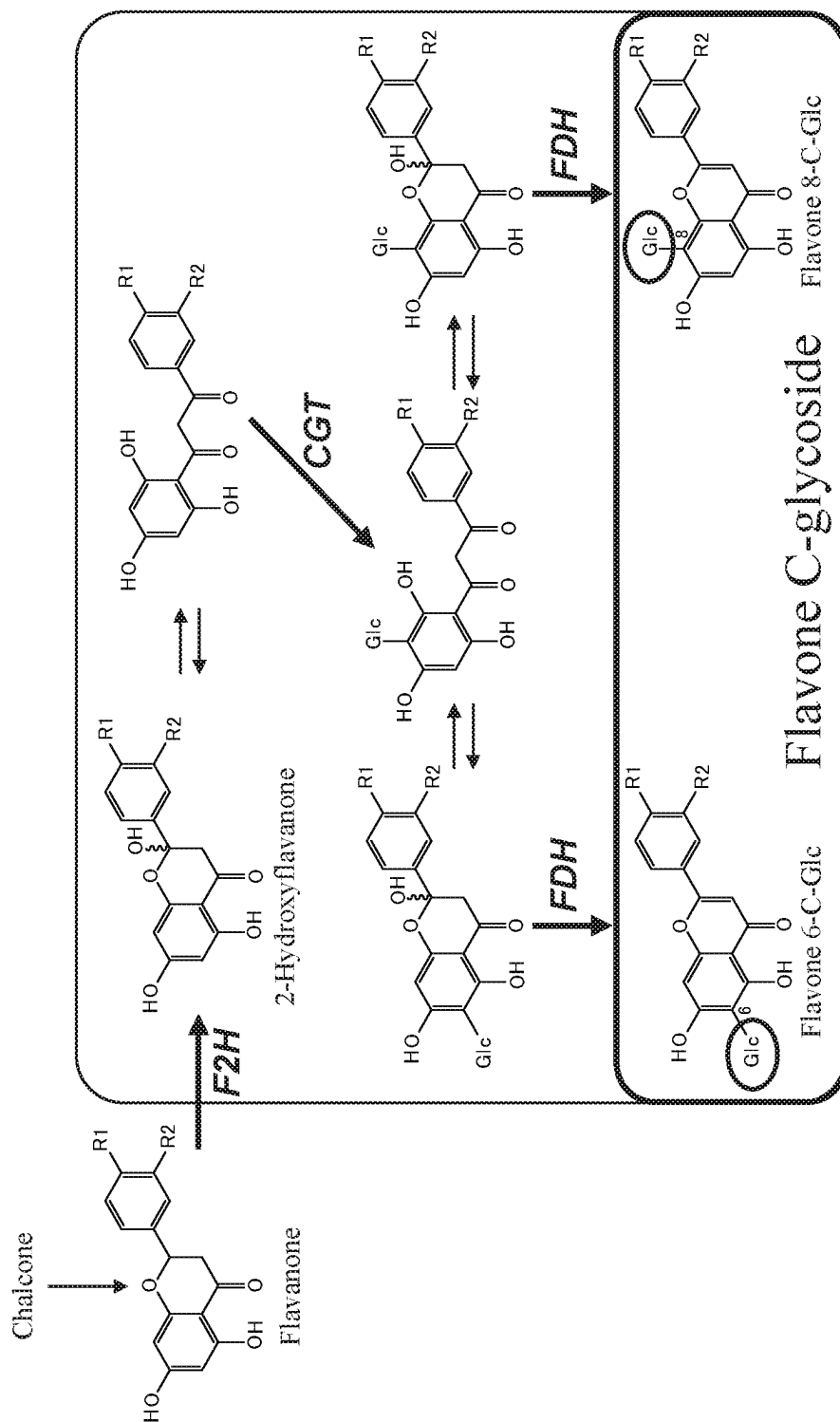


FIG. 2

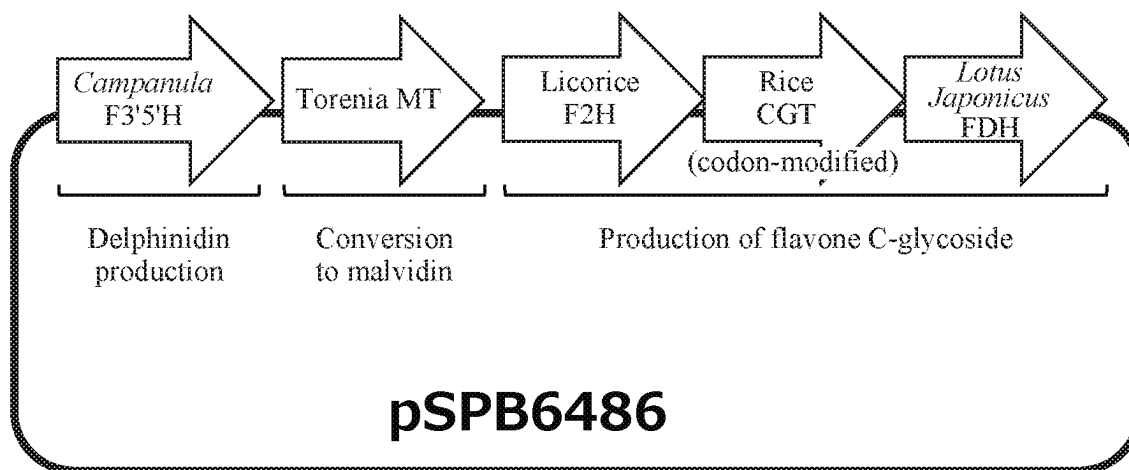


FIG. 3

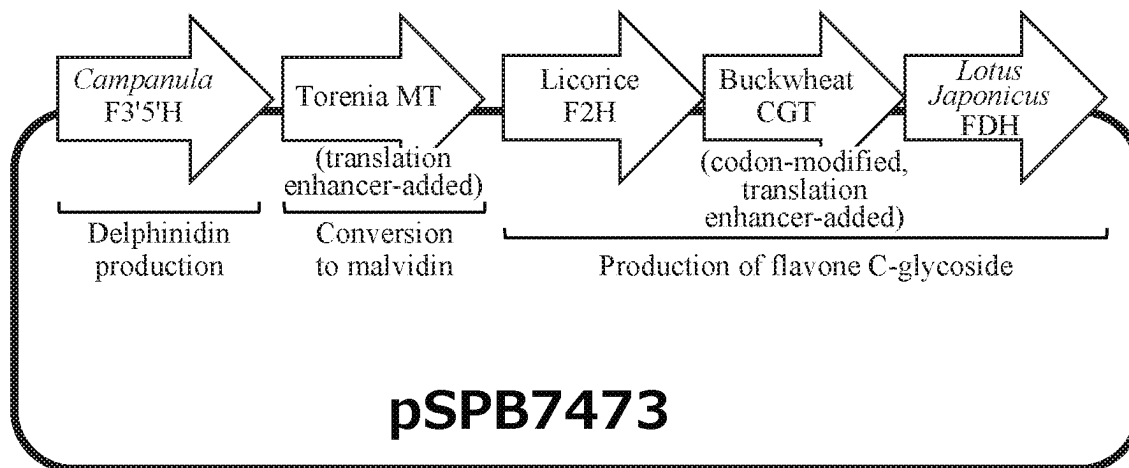


FIG. 4

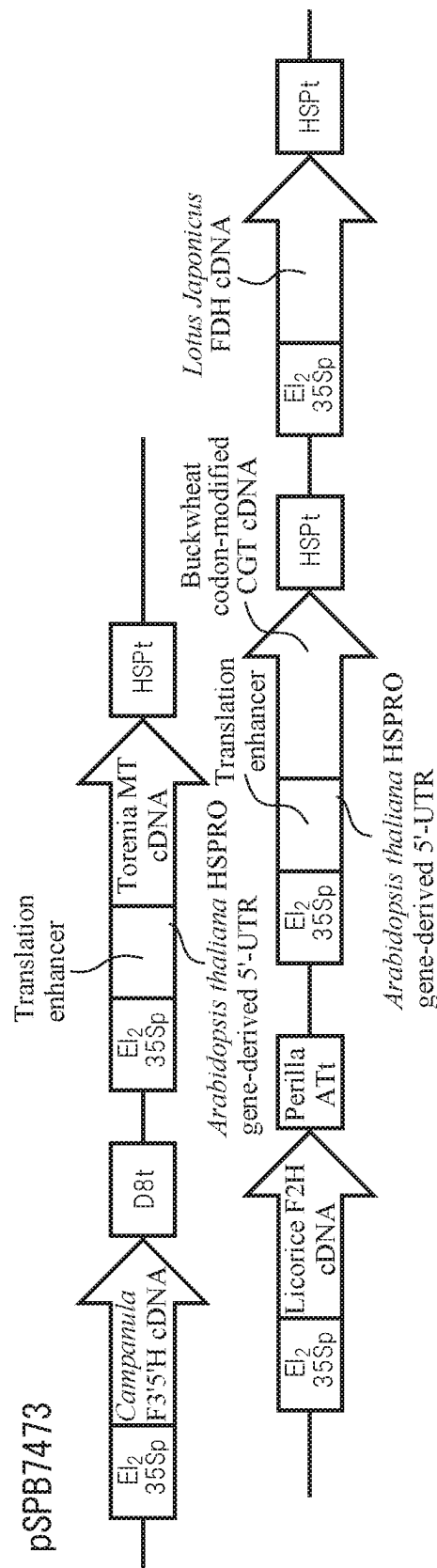


FIG. 5

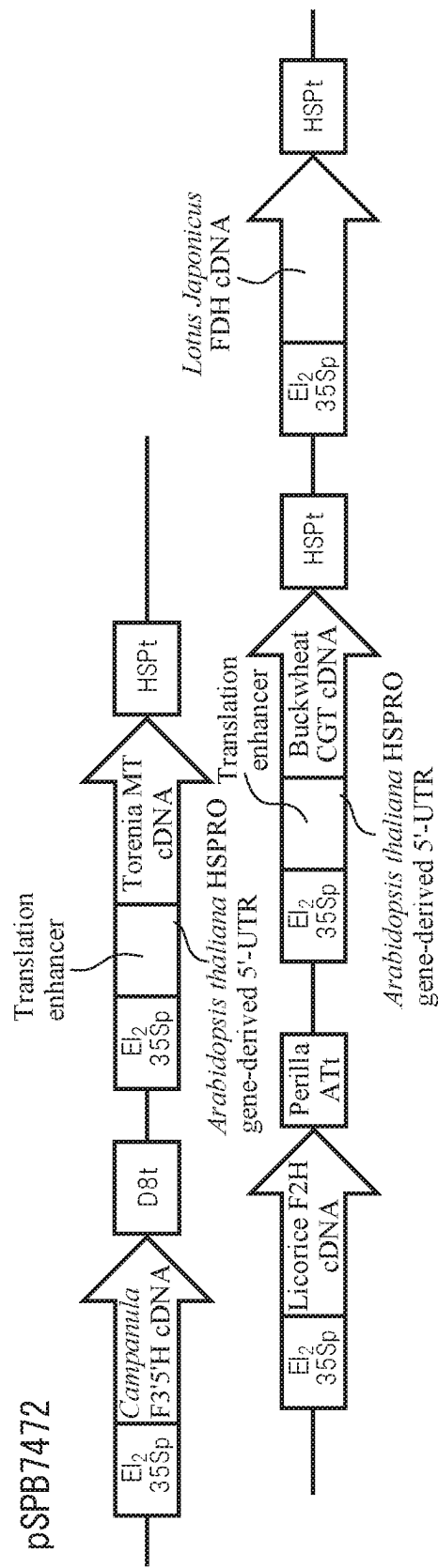


FIG. 6

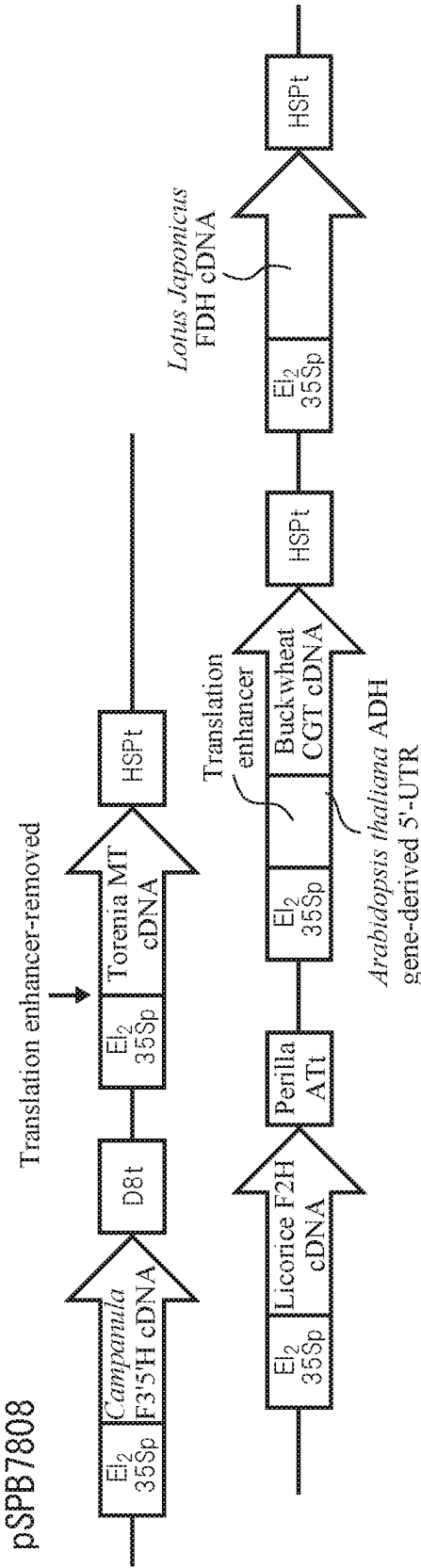
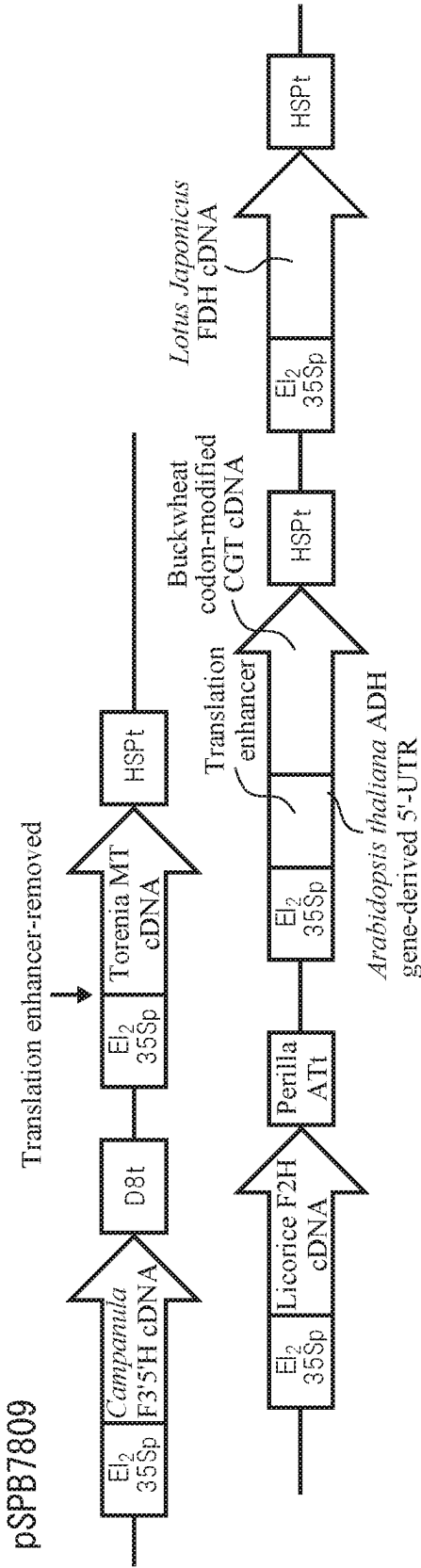


FIG. 7



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BUCKWHEAT-DERIVED C-GLUCOSYLTRANSFERASE GENE AND UTILIZATION THEREOF

FIELD

The present invention relates to buckwheat-derived C-glucosyltransferase (CGT) genes or their homologs, and to a method for creating transgenic plants with blue flower color, comprising a step of using the genes to cause delphinidin-type anthocyanins and flavone mono-C-glycosides to be copresent in plant cells.

BACKGROUND

Rose, chrysanthemum and carnation are industrially important ornamental flowers worldwide. Rose in particular, being the most popular flowering plant, has a record of cultivation since ancient times, and it has been artificially crossbred for hundreds of years. One problem, however, has been that none of the hybridizable related species have wild varieties with blue flower color, and it has therefore been difficult to create rose varieties with blue flower color by conventional cross-breeding and mutation breeding. Creating completely new blue flower colors should lead to new demand for even wider uses of ornamental flowers, and should help to increase production and consumption. It has therefore been attempted to create roses with blue flower colors by genetic engineering methods.

Flowers with purple to blue colors, for example, are known to abundantly contain delphinidin-type anthocyanins having delphinidin, petunidin and malvidin backbones, but since ornamental flowers such as rose cannot produce such delphinidin-type anthocyanins, research continues to be conducted with the aim of artificially producing delphinidins by expressing the flavonoid 3',5'-hydroxylase gene that is necessary for their synthesis (NPL 1). However, even when plant metabolism is artificially modified in order to express an enzyme gene that produces a substance of interest in the recombinant plant, often little or absolutely none of the substance of interest accumulates.

Moreover, the color of a flower changes not only by the structures of the anthocyanins themselves as the essential pigments, but also due to copresent flavonoids (also known as copigments), metal ions, and the vacuole pH. Flavones or flavonols are typical copigments that form sandwich-like layers with anthocyanins and render the anthocyanins blue, producing a deepening effect (NPL 2). This is known as the "copigment effect". Flavones, in particular, are known to exhibit a powerful copigment effect, and analysis of gene recombinant carnations, for example, has demonstrated that flavones exhibit a significant copigment effect (NPL 3). For Dutch iris, it has been reported that a higher ratio of the total flavone content with respect to the total delphinidin content results in a more powerful copigment effect, and a bluer color (NPL 4).

However, not all plants can produce flavones, and it is known that roses and petunias do not store flavones. Attempts have therefore been made to modify flower color by expressing in the plants different genes coding for proteins having activity for synthesizing flavones from flavanones (PTL 1).

In plants, flavones are distributed not only in free form but also as glycosides, with flavone O-glycosides and flavone C-glycosides being formed primarily, and flavone C-glycosides being known to exhibit a particularly powerful copigment effect. For example, isovitexin, as one type of flavone

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C-glycoside, has been reported to exhibit a copigment effect with anthocyanins in Japanese garden iris (*Iris ensata* Thunb.), and to produce a blue flower color via stabilization of anthocyanins (NPL 5). One known biosynthetic pathway for flavone C-glycosides is synthesis from flavanones via reaction catalyzed by flavanone 2-hydroxylase (F2H), C-glucosyltransferase (CGT) and dehydratase (FDH) (NPL 6).

Previously it has been reported that roses with blue flower color were created by introducing the *Campanula*-derived F3',5'H gene and the *torenia*-derived MT gene, licorice-derived F2H gene, rice-derived CGT gene and *Lotus japonicus*-derived FDH gene, to cause copresence of delphinidin-type anthocyanins and flavone C-glycosides in plant cells (PTL 2). However, the flower colors of rose varieties created in this manner are found to have a strong reddish tint, and it is therefore still desired to develop techniques for controlling blue color expression in order to allow uniform and stable creation of roses with bluer flower colors.

CITATION LIST

Patent Literature

- [PTL 1] Japanese Unexamined Patent Publication No. 2000-279182
- [PTL 2] International Patent Publication No. 2019/069946
- [PTL 3] International Patent Publication No. 2008/156206

Non Patent Literature

- [NPL 1] Phytochemistry Reviews 5, 283-291
- [NPL 2] Prog. Chem. Org. Natl. Prod. 52
- [NPL 3] Phytochemistry, 63, 15-23(2003)
- [NPL 4] Plant Physiol. Bioch. 72, 116-124(2013)
- [NPL 5] Euphytica 115, 1-5(2000)
- [NPL 6] FEBS Lett. 589, 182-187(2015)

SUMMARY

Technical Problem

The problem to be solved by the invention is to create a transgenic plant having a uniform and stable blue flower color (RHS Color Chart 5th Edition: Violet-Blue group/Blue group and/or hue angle: 339.7°-270.0°), based on research on the causes of redness in flower color.

Solution to Problem

As a result of ardent research and much experimentation on this problem, the present inventors have found that flavone di-C-glycosides are a cause of redness, and that the copigment effect of flavone mono-C-glycosides with delphinidin-type anthocyanins is significantly higher than that of flavone di-C-glycosides. By transferring the buckwheat-derived CGT gene, which was selected from among various CGT genes, the present inventors further succeeded in significantly accumulating only flavone mono-C-glycosides in plant petals. The invention has been completed upon these findings.

Specifically, the present invention provides the following.

- [1] A buckwheat-derived CGT gene or its homolog, wherein:
the buckwheat-derived CGT gene or its homolog is selected from the group consisting of:

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(1-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 11;

(1-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 11, under stringent conditions, and has the same activity as the polynucleotide of (1-a); and

(1-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 12.

[2] The buckwheat-derived CGT gene or its homolog according to [1], to which the *Arabidopsis thaliana* ADH gene-derived untranslated region (5'-UTR) (SEQ ID NO: 15) or the *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) has been added.

[3] A vector comprising a buckwheat-derived CGT gene or its homolog, wherein the buckwheat-derived CGT gene or its homolog is selected from the group consisting of:

(1-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 11;

(1-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 11 under stringent conditions and has the same activity as the polynucleotide of (1-a);

(1-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 12;

(1-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 12 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (1-c); and

(1-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 12 and having the same activity as a protein encoded by the polynucleotide of (1-c).

[4] The vector according to [3], wherein the *Arabidopsis thaliana* ADH gene-derived untranslated region (5'-UTR) (SEQ ID NO: 15) or the *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) has been added to the buckwheat-derived CGT gene or its homolog.

[5] The vector according to [3] or [4], which further comprises a flavanone 2-hydroxylase (F2H) gene or its homolog, and a dehydratase (FDH) gene or its homolog.

[6] The vector according to [5], wherein the F2H gene or its homolog is selected from the group consisting of:

(2-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 5;

(2-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 5 under stringent conditions and has the same activity as the polynucleotide of (2-a);

(2-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 6;

(2-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 6 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (2-c); and

(2-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect

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to the amino acid sequence listed as SEQ ID NO: 6 and having the same activity as a protein encoded by the polynucleotide of (2-c), and

the FDH gene or its homolog is selected from the group consisting of:

(3-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 9;

(3-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 9 under stringent conditions and has the same activity as the polynucleotide of (3-a);

(3-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 10;

(3-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 10 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (3-c); and

(3-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 10 and having the same activity as a protein encoded by the polynucleotide of (3-c).

[7] The vector according to any one of [3] to [6], which further includes a flavonoid F3',5' hydroxylase (F3',5'H) gene or its homolog, and a methyltransferase (MT) gene or its homolog.

[8] The vector according to [7], wherein the F3',5'H gene or its homolog is selected from the group consisting of:

(4-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 1;

(4-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 1 under stringent conditions and has the same activity as the polynucleotide of (4-a);

(4-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 2;

(4-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 2 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (4-c); and

(4-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 2 and having the same activity as a protein encoded by the polynucleotide of (4-c), and

the MT gene or its homolog is selected from the group consisting of:

(5-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 3;

(5-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 3 under stringent conditions and has the same activity as the polynucleotide of (5-a);

(5-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 4;

(5-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 4 having a deletion, substitution,

insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (5-c); and

(5-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 4 and having the same activity as a protein encoded by the polynucleotide of (5-c).

[9] A transgenic plant comprising a buckwheat-derived CGT gene or its homolog according to [1] or [2] or a vector according to any one of [3] to [8], or its inbred or outbred progeny.

[10] The transgenic plant according to [9], or its inbred or outbred progeny, wherein the plant is selected from among rose, chrysanthemum, carnation or lily.

[11] The transgenic plant according to [10], or its inbred or outbred progeny, wherein the plant is rose.

[12] Propagules, partial plant bodies, tissue or cells of a transgenic plant according to any one of [9] to [11] or its inbred or outbred progeny.

[13] Cut flowers of a transgenic plant according to any one of [9] to [11], or its inbred or outbred progeny, or a processed form created from the cut flowers.

[14] A method for creating transgenic plants with blue flower color, comprising a step of transferring a buckwheat-derived C-glucosyltransferase (CGT) gene or its homolog into a host plant to cause delphinidin-type anthocyanins and flavone mono-C-glycosides to coexist in the plant cells.

[15] The method according to [14], wherein the flavone mono-C-glycoside is apigenin 6-C-glucoside, luteolin 6-C-glucoside, tricetin 6-C-glucoside, apigenin 8-C-glucoside, luteolin 8-C-glucoside or tricetin 8-C-glucoside, or a derivative thereof.

[16] The method according to [14] or [15], wherein the delphinidin-type anthocyanin is selected from the group consisting of malvidins, delphinidins, petunidins and their combinations.

[17] The method according to any one of [14] to [16], which further comprises transferring a buckwheat-derived CGT gene or its homolog according to [1] or [2] or a vector according to any one of [3] to [8] into host plant cells.

[18] The method according to any one of [14] to [17], wherein the plant is rose, chrysanthemum, carnation or lily.

[19] The method according to [18], wherein the plant is rose.

Advantageous Effects of Invention

According to the invention it is possible to uniformly and stably create a plant variety having a blue flower color (RHS Color Chart 5th Edition: Violet-Blue group/Blue group and/or hue angle: 339.7°-270.0°).

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 shows the biosynthetic pathway for a flavone mono-C-glycoside in a plant.

FIG. 2 shows the structure of pSPB6486.

FIG. 3 shows the structure of pSPB7473.

FIG. 4 shows the detailed structure of pSPB7473.

FIG. 5 shows the detailed structure of pSPB7472.

FIG. 6 shows the detailed structure of pSPB7808.

FIG. 7 shows the detailed structure of pSPB7809.

Anthocyanins are a group of pigments that are widely extant in plants, and they are known to exhibit red, blue and purple flower colors. They are classified into 3 types, pelargonidin, cyanidin and delphinidin, based on the number of

hydroxyl groups on the B-ring of the anthocyanidin, as the aglycone form. The chromophoric group is the aglycone portion, with pelargonidin-type anthocyanins exhibiting orange color, cyanidin-type anthocyanins exhibiting red color and delphinidin-type anthocyanins exhibiting purple to blue color. Throughout the present specification, "delphinidin-type anthocyanins" also include their derivatives having delphinidin, malvidin or petunidin backbones, with malvidin being preferred.

When delphinidin-type anthocyanins are copresent with substances such as flavones, flavonols, organic acid esters and tannins, their molecular interaction often produces blueish colors. This phenomenon is known as "copigmentation", and substances that produce the phenomenon are known as copigments. Copigmentation includes not only a color depth effect that induces blue color production, but also a deep color effect or an effect of increasing color stability. The present inventors have confirmed that copigmentation between delphinidin-type anthocyanins and flavone C-glycosides causes blue color expression in rose petals (PTL 2).

Flavones are organic compounds that are flavan-derived cyclic ketones, and in plants they mainly exist as glycosides. Flavone, in the strict definition, refers to 2,3-dihydroflavan-4-one, which is a compound with chemical formula $C_{15}H_{10}O_2$ and molecular weight 222.24, but in the wider sense flavones are a category of flavonoids, a flavonoid being classified as a "flavone" if it has a flavone structure as the basic backbone and also lacks the hydroxyl group at the 3-position. As used herein, "flavone C-glycoside" means a glycoside of a flavone in the wide sense, i.e. a derivative falling under the definition of flavones, wherein an aglycone is directly bonded to the anomeric carbon of an aldose. Flavone C-glycosides include, but are not limited to, luteolin C-glycoside, tricetin C-glycoside, apigenin C-glycoside and acacetin C-glycoside.

Flavone C-glycosides also include glycosides of apigenin, luteolin, tricetin and acacetin derivatives. One pathway known for the biosynthetic pathway of flavone C-glycosides in plants is the pathway shown in FIG. 1. In this pathway, flavone C-glycoside is produced via F2H, CGT and FDH.

In this synthesis pathway, flavone di-C-glycosides such as vicianin-2 (apigenin 6,8-di-C-glucoside) are also synthesized in addition to flavone mono-C-glycosides, but the present inventors have found, surprisingly, that flavone di-C-glycosides are a cause of redness, and that flavone mono-C-glycosides have a higher copigment effect with delphinidin-type anthocyanins than flavone di-C-glycosides. In order to uniformly and stably create a transgenic plant with blue flower color, it is necessary to reduce to a minimum the accumulation of flavone di-C-glycosides, and to accumulate significant amounts of flavone mono-C-glycosides alone in the petals. Flavone mono-C-glycosides are typically flavone 6-C-glucosides or flavone 8-C-glucosides, but are preferably flavone 6-C-glucosides. Examples of flavone mono-C-glycosides include apigenin 6-C-glucoside (isovitexin), luteolin 6-C-glucoside (isoorientin), tricetin 6-C-glucoside, apigenin 8-C-glucoside (vitexin), luteolin 8-C-glucoside (orientin) and tricetin 8-C-glucoside, or their derivatives.

Accumulation of flavone C-glycosides in plant cells can be achieved by transforming host plants with vectors comprising genes necessary for the aforementioned synthesis pathways (i.e. the F2H gene, CGT gene and FDH gene), or their homologs. If the CGT gene used is a buckwheat-derived CGT gene or its homolog, and particularly a buckwheat-derived CGT gene or its homolog with the *Arabidopsis thaliana* ADH gene-derived untranslated region (5'-

UTR) (SEQ ID NO: 15) or *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) added as a translation enhancing sequence, it is possible to accumulate significantly more flavone mono-C-glycosides in petals than flavone di-C-glycosides.

The buckwheat-derived CGT gene or its homolog is selected from the group consisting of the following polynucleotides:

(1-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 11;

(1-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 11 under stringent conditions and has the same activity as the polynucleotide of (1-a);

(1-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 12;

(1-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 12 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (1-c); and

(1-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 12 and having the same activity as a protein encoded by the polynucleotide of (1-c).

The source of the F2H gene or its homolog is not particularly restricted so long as it has the desired function, but it is preferably a licorice-derived F2H gene or its homolog, and selected from the group consisting of the following polynucleotides:

(2-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 5;

(2-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 5 under stringent conditions and has the same activity as the polynucleotide of (2-a);

(2-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 6;

(2-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 6 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (2-c); and

(2-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 6 and having the same activity as a protein encoded by the polynucleotide of (2-c).

The source of the FDH gene or its homolog is not particularly restricted so long as it has the desired function, but it is preferably a *Lotus japonicus*-derived FDH gene or its homolog, and selected from the group consisting of the following polynucleotides:

(3-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 9;

(3-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 9 under stringent conditions and has the same activity as the polynucleotide of (3-a);

(3-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 10;

(3-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 10 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (3-c); and

(3-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 10 and having the same activity as a protein encoded by the polynucleotide of (3-c).

Accumulation of delphinidin-type anthocyanins in plant cells can be achieved by incorporating a flavonoid F3',5'-hydroxylase (F3',5'H) gene or its homolog and a methyltransferase (MT) gene or its homolog in a host plant (PTL 3). By transforming a host plant with a vector further comprising a F3',5'H gene or its homolog and an MT gene or its homolog in addition to a gene necessary for the synthesis pathway of the aforementioned flavone mono-C-glycosides or their homologs, it is possible to cause a delphinidin-type anthocyanin and a flavone mono-C-glycoside to coexist in the host plant cells.

The source of the F3',5'H gene or its homolog is not particularly restricted so long as it has the desired function, but it is preferably a *Campanula*-derived F3',5'H gene or its homolog, and selected from the group consisting of:

(4-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 1;

(4-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 1 under stringent conditions and has the same activity as the polynucleotide of (4-a);

(4-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 2;

(4-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 2 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (4-c); and

(4-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect to the amino acid sequence listed as SEQ ID NO: 2 and having the same activity as a protein encoded by the polynucleotide of (4-c).

The source of the MT gene or its homolog is not particularly restricted so long as it has the desired function, but it is preferably a *torenia*-derived MT gene or its homolog, and selected from the group consisting of:

(5-a) a polynucleotide consisting of the nucleotide sequence listed as SEQ ID NO: 3;

(5-b) a polynucleotide that hybridizes with a polynucleotide consisting of the nucleotide sequence complementary to the nucleotide sequence listed as SEQ ID NO: 3 under stringent conditions and has the same activity as the polynucleotide of (5-a);

(5-c) a polynucleotide that encodes a protein consisting of the amino acid sequence listed as SEQ ID NO: 4;

(5-d) a polynucleotide that encodes a protein consisting of an amino acid sequence that is the amino acid sequence listed as SEQ ID NO: 4 having a deletion, substitution, insertion and/or addition of one or more amino acids, and having the same activity as a protein encoded by the polynucleotide of (5-c); and

(5-e) a polynucleotide that encodes a protein having an amino acid sequence with at least 90% identity with respect

to the amino acid sequence listed as SEQ ID NO: 4 and having the same activity as a protein encoded by the polynucleotide of (5-c).

The *Arabidopsis thaliana* ADH gene-derived untranslated region (5'-UTR) (SEQ ID NO: 15) or the *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) may also be added as a translation enhancing sequence to the MT gene or its homolog.

Throughout the present specification, the term "polynucleotide" refers to DNA or RNA.

As used herein, the term "stringent conditions" refers to conditions that allow specific binding between a polynucleotide or oligonucleotide and genomic DNA in a selective and detectable manner. Stringent conditions are defined by an appropriate combination of salt concentration, organic solvent (for example, formamide), temperature and other known conditions. Specifically, stringency is increased by reducing the salt concentration, increasing the organic solvent concentration or raising the hybridization temperature. Stringency is also affected by the rinsing conditions after hybridization. The rinsing conditions are defined by the salt concentration and temperature, and stringency of rinsing is increased by reducing the salt concentration and raising the temperature. Therefore, the term "stringent conditions" means conditions such that specific hybridization takes place only between nucleotide sequences with high identity, such as a degree of "identity" between the nucleotide sequences of about 80% or greater, preferably about 90% or greater, more preferably about 95% or greater, even more preferably 97% or greater and most preferably 98% or greater, on average. The "stringent conditions" may be, for example, a temperature of 60° C. to 68° C., a sodium concentration of 150 to 900 mM and preferably 600 to 900 mM, and a pH of 6 to 8, with specific examples including hybridization under conditions of 5×SSC (750 mM NaCl, 75 mM trisodium citrate), 1% SDS, 5×Denhardt solution, 50% formaldehyde, 42° C., and rinsing under conditions of 0.1×SSC (15 mM NaCl, 1.5 mM trisodium citrate), 0.1% SDS, 55° C.

The hybridization may be carried out by a method that is publicly known in the field or a similar method, such as the method described in Current Protocols in Molecular Biology (edited by Frederick M. Ausubel et al., 1987). When a commercially available library is to be used, the hybridization may be carried out according to the method described in the accompanying directions for use. The gene selected by hybridization may be naturally derived, such as plant-derived or non-plant-derived. The gene selected by the hybridization may be cDNA, genomic DNA or chemically synthesized DNA.

The phrase "amino acid sequence having a deletion, substitution, insertion and/or addition of one or more amino acids", as used herein, means an amino acid sequence having a deletion, substitution, insertion and/or addition of 1 to 20, preferably 1 to 5 and more preferably 1 to 3 arbitrary amino acids. Site-specific mutagenesis is a useful genetic engineering method as it allows introduction of specific mutations into specified sites, and it may be carried out by the method described in Molecular Cloning: A Laboratory Manual, 2nd Ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y., 1989. By expressing the mutant DNA using a suitable expression system, it is possible to obtain a protein consisting of an amino acid sequence having a deletion, substitution, insertion and/or addition of one or more amino acids.

A polynucleotide can be obtained by a method that is publicly known to those skilled in the art, such as a method of chemical synthesis using the phosphoramidite method, or

a nucleic acid amplification method using a plant nucleic acid specimen as template, and primers designed based on the nucleotide sequence of the target gene.

Throughout the present specification, the term "identity" means, for polypeptide sequences (or amino acid sequences) or polynucleotide sequences (or nucleotide sequences), the quantity (number) of amino acid residues or nucleotides composing them that can be determined to be identical between the two chains, in the sense of mutual agreement between them, meaning the degree of sequence correlation between two polypeptide sequences or two polynucleotide sequences, and this "identity" can be easily calculated. Numerous methods are known for measuring identity between two polynucleotide sequences or polypeptide sequences, and the term "identity" is well known to those skilled in the art (for example, see Lesk, A. M. (Ed.), Computational Molecular Biology, Oxford University Press, New York, (1988); Smith, D. W. (Ed.), Biocomputing: Informatics and Genome Projects, Academic Press, New York, (1993); Griffin, A. M. & Griffin, H. G. (Ed.), Computer Analysis of Sequence Data: Part I, Human Press, New Jersey, (1994); von Heinje, G., Sequence Analysis in Molecular Biology, Academic Press, New York, (1987); Gribskov, M. & Devereux, J. (Ed.), Sequence Analysis Primer, M-Stockton Press, New York, (1991) and elsewhere).

Also, the numerical values for "identity" used in the present specification, unless otherwise specified, may be the numerical values calculated using an identity search program known to those skilled in the art, but they are preferably numerical values calculated using the ClustalW program of MacVector Application (version 9.5, Oxford Molecular Ltd., Oxford, England). According to the invention, the degree of "identity" between polynucleotide sequences or amino acid sequences is, for example, about 90% or greater, preferably about 95% or greater, more preferably about 97% or greater, and most preferably about 98% or greater.

The polynucleotide (nucleic acid, gene) of the invention "encodes" a protein of interest. Here, "encodes" means that it allows expression of the protein of interest in a state in which it exhibits its activity. Also, the term "encodes" includes both encoding a structural sequence (exon) that is a continuous section of the protein of interest, and encoding via an intervening sequence (intron).

A gene with a natural nucleotide sequence can be obtained by analysis using a DNA sequencer, for example. Also, DNA encoding an enzyme having a modified amino acid sequence can be synthesized using common site-specific mutagenesis or PCR, based on DNA having the natural nucleotide sequence. For example, a DNA fragment to be modified may be obtained by restriction enzyme treatment of natural cDNA or genomic DNA, and used as template for site-specific mutagenesis or PCR using primers with the desired mutation, to obtain a DNA fragment having the desired modification. The DNA fragment having the mutation may then be linked with a DNA fragment encoding another portion of the target enzyme.

Alternatively, in order to obtain DNA encoding an enzyme consisting of a shortened amino acid sequence, DNA encoding an amino acid sequence longer than the target amino acid sequence, such as the full-length amino acid sequence, may be cut with a desired restriction enzyme, and if the obtained DNA fragment does not code for the full target amino acid sequence, then a DNA fragment consisting of the sequence of the missing portion may be synthesized and linked to it.

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By expressing the obtained polynucleotide using a gene expression system in *Escherichia coli* or yeast and measuring the enzyme activity, it is possible to confirm that the obtained polynucleotide encodes a protein with the desired activity.

The present invention relates to a (recombinant) vector, and especially an expression vector, including the aforementioned polynucleotide, and to chrysanthemum plants transformed by the vector.

The vector of the invention also comprises an expression control region, such as a promoter, terminator and replication origin, that are dependent on the type of host plant into which it is introduced. Examples of promoters that constitutively express polynucleotides in plant cells include cauliflower mosaic virus 35S promoter, EL₂35S promoter having two 35S promoter enhancer regions linked together, and the rd29A gene promoter, rbcS promoter and mac-1 promoter. For tissue-specific gene expression, a promoter for a gene expressed specifically in that tissue may be used.

The vector may be created by a common method using a restriction enzyme and ligase. Transformation of a host plant using the expression vector may also be carried out by a common method.

At the current level of technology, it is possible to use techniques to introduce a polynucleotide into a plant and constitutively or tissue-specifically express the polynucleotide. Transfer of the DNA into the plant may be carried out by a method known to those skilled in the art, such as the *Agrobacterium* method, binary vector method, electroporation method, PEG method or particle gun method.

Plants to be used as hosts for the invention are not particularly restricted and may be plants belonging to genus Rosaceae *Rosa*, Compositae *Chrysanthemum*, Caryophyllaceae *Dianthus* (such as carnation) or Liliaceae *Lilium*, among which rose cultivar of Rosaceae *Rosa* (scientific name: *Rosa hybrida*) is especially preferred. The term "rose plant", as used herein, is a rose cultivar of Rosaceae *Rosa* (scientific name: *Rosa hybrida*), which is its taxonomical classification. Roses are largely classified as Hybrid Tea, *Floribunda* and *Polyantha* roses based on their tree form and flower size, with the major pigment (anthocyanin) in the petals of all lines being of two types, the cyanidin-type and pelargonidin-type. The type of rose plant used as a host for the invention is not particularly restricted, and any of these varieties or lines are suitable. Examples of rose varieties to be used as hosts include Ocean Song, Noblesse, Rita Perfumera, Cool Water, Fame, Topless and Peach Avalanche.

By means of the present invention it is possible to uniformly and stably create transgenic plants, preferably Rosaceae *Rosa*, Compositae *Chrysanthemum* and Caryophyllaceae *Dianthus* (carnation), and most preferably rose plants, having blue flower colors. When the obtained transgenic plant is a rose plant, it exhibits a flower color in the Blue group or Violet-Blue group according to the RHS Color Chart, and/or with a hue angle of 339.7° to 270.0°, and more preferably 315° or smaller, in the CIEL*a*b* color system.

The invention still further relates to cut flowers of the obtained transgenic plant or its inbred or outbred progeny, or the propagules, partial plant body, tissue or cells, or a processed form created from the cut flowers (especially processed cut flowers). The processed cut flowers referred to here include pressed flowers formed using cut flowers, or preserved flowers, dry flowers or resin sealed products, with no limitation to these.

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The present invention will now be explained in greater detail by examples.

EXAMPLES

Example 1: Simulation of Flavone C-Glycoside Copigment Effect with Anthocyanin (Malvin)

An anthocyanin (malvin) and flavone C-glycoside were prepared to simulate the copigment effect of the flavone C-glycoside on malvin. The malvin (malvidin 3,5-diglucoside) and flavone C-glycosides (vitexin (apigenin 8-C-glucoside), isovitexin (apigenin 6-C-glucoside), orientin (luteolin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside) and vicienin-2 (apigenin 6,8-di-C-glucoside)) used in the experiments were purchased from Nacalai Tesque, Inc.

Each flavone C-glycoside (vitexin, isovitexin, orientin, isoorientin or vicienin-2) was added to the acquired malvin at a 5-molar equivalent concentration in a buffering solution at pH 5.0, and the absorption spectra were measured. The malvin concentration was 0.5 mM.

TABLE 1

Absorption maximum (λ _{max}) and hue angle (°) of malvin solution upon flavone C-glycoside addition			
		Absorption maximum (λ _{max})	Hue angle
Isovitexin (apigenin 6-C-glucoside)	5 equiv.	576 nm	298°
Vitexin (apigenin 8-C-glucoside)	5 equiv.	563 nm	312°
Vicienin-2 (apigenin 6,8-di-C-glucoside)	5 equiv.	562 nm	314°
Isoorientin (luteolin 6-C-glucoside)	5 equiv.	575 nm	299°
Orientin (luteolin 8-C-glucoside)	5 equiv.	570 nm	307°
—	—	531 nm	334°

Addition of a flavone C-glycoside increased the absorbance of the malvin solution and shifted the absorption maximum (λ_{max}) toward the long wavelength end compared to malvin alone, regardless of which flavone C-glycoside was added. This effect was confirmed as a greater shift in absorption maximum toward the long wavelength end in the order: isovitexin > isoorientin > orientin > vitexin > vicienin-2. The hue angle was also confirmed to be smaller in the order: isovitexin > isoorientin > orientin > vitexin > vicienin-2. It was thus demonstrated that the copigment effect of flavone mono-C-glycosides is higher than that of flavone di-C-glycosides. In addition, it was shown that the copigment effect of flavone mono-C-glycosides is particularly high with flavone 6-C-glycosides.

Example 2: Transfer of *Campanula*-Derived F3',5'H Gene, *Torenia*-Derived MT Gene, *Licorice*-Derived F2H Gene, Rice-Derived Codon Usage-Modified CGT Gene and *Lotus japonicus*-Derived FDH Gene into Rose Variety "Ocean Song"

Plasmid pSPB6486 has pBINPLUS as the basic backbone, and contains the following four expression cassettes.

- (1) EL₂35S promoter, *Campanula*-derived F3',5'H full-length cDNA (SEQ ID NO: 1) and D8 terminator
- (2) EL₂35S promoter, *torenia*-derived MT full-length cDNA (SEQ ID NO: 3) and NOS terminator
- (3) 35S promoter, *licorice*-derived F2H full-length cDNA (SEQ ID NO: 5) and *perilla*-derived AT terminator

(4) 35S promoter, rice-derived codon usage-modified CGT full-length cDNA (SEQ ID NO: 7) and *Arabidopsis thaliana*-derived HSP terminator

(5) 35S promoter, *Lotus japonicus*-derived FDH full-length cDNA (SEQ ID NO: 9) and *Arabidopsis thaliana*-derived HSP terminator

order: vicenin-2>isovitexin >vitexin >isorientin >orientin, with a maximum total amount of 1.563 mg per 1 g of fresh petal weight. The total amount of flavone C-glycosides was more than about 10 times that of malvidins.

The measured values for representative transformants are shown in Table 2 below.

TABLE 2

Plant	Mal	Anthocyanidin (mg/g)				Flavonol (mg/g)			Flavone (mg/g)			Flavone C-glycoside (mg/g)					
No.	(%)	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori
Host	0.0	0.000	0.024	0.000	0.000	0.000	0.000	2.883	0.586	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1	70.8	0.024	0.000	0.009	0.000	0.080	1.225	0.077	0.023	0.000	0.013	0.010	0.187	0.205	0.505	0.009	0.073
2	66.2	0.025	0.000	0.007	0.000	0.062	2.453	0.181	0.020	0.009	0.047	0.029	0.743	0.109	0.319	0.027	0.077
3	54.6	0.022	0.006	0.008	0.000	0.050	1.058	1.381	0.515	0.000	0.000	0.000	0.896	0.093	0.219	0.000	0.015
4	52.7	0.020	0.003	0.007	0.000	0.035	0.858	0.482	0.178	0.000	0.003	0.006	0.429	0.174	0.354	0.007	0.035
5	64.0	0.028	0.000	0.009	0.000	0.065	2.045	0.168	0.063	0.000	0.000	0.000	0.586	0.130	0.299	0.002	0.030
6	67.7	0.028	0.000	0.007	0.000	0.074	1.593	0.455	0.115	0.011	0.010	0.009	0.795	0.126	0.355	0.018	0.050
7	72.1	0.027	0.000	0.008	0.000	0.089	2.753	0.526	0.183	0.006	0.004	0.004	0.905	0.093	0.262	0.000	0.028
8	54.7	0.019	0.007	0.005	0.000	0.049	1.010	2.128	0.418	0.018	0.002	0.005	0.945	0.226	0.325	0.000	0.021
9	39.1	0.018	0.009	0.004	0.000	0.025	0.679	1.438	0.322	0.005	0.002	0.002	0.672	0.246	0.283	0.007	0.020
10	39.8	0.013	0.009	0.005	0.000	0.021	0.447	1.129	0.586	0.000	0.000	0.002	0.432	0.080	0.179	0.000	0.010
11	45.9	0.011	0.011	0.006	0.000	0.034	0.914	3.471	1.481	0.000	0.000	0.000	0.839	0.050	0.118	0.000	0.003
12	45.4	0.022	0.005	0.007	0.000	0.032	0.803	0.992	0.418	0.000	0.004	0.005	0.511	0.142	0.354	0.000	0.024
13	62.8	0.032	0.000	0.008	0.000	0.067	1.830	0.210	0.059	0.018	0.023	0.015	0.910	0.000	0.249	0.000	0.049
14	65.6	0.024	0.000	0.007	0.000	0.060	1.942	0.258	0.078	0.011	0.019	0.015	1.047	0.056	0.305	0.016	0.054
15	52.9	0.042	0.002	0.009	0.000	0.062	0.993	0.377	0.100	0.000	0.004	0.005	0.396	0.197	0.450	0.000	0.050
16	63.9	0.031	0.000	0.007	0.000	0.067	2.742	0.221	0.056	0.009	0.030	0.018	0.591	0.063	0.285	0.022	0.058
17	63.5	0.016	0.000	0.005	0.000	0.037	1.767	0.102	0.013	0.009	0.051	0.029	0.621	0.147	0.242	0.022	0.061
18	74.5	0.029	0.000	0.009	0.000	0.109	2.078	0.187	0.029	0.017	0.030	0.015	0.604	0.051	0.276	0.021	0.061
19	55.4	0.019	0.009	0.006	0.000	0.059	0.937	1.893	0.502	0.008	0.001	0.000	1.002	0.078	0.218	0.000	0.007
20	38.5	0.006	0.010	0.002	0.000	0.019	0.480	2.621	0.468	0.007	0.001	0.003	0.890	0.147	0.247	0.000	0.012
21	47.7	0.025	0.000	0.009	0.000	0.031	2.684	0.376	0.102	0.008	0.019	0.018	0.700	0.147	0.303	0.017	0.048
22	39.5	0.011	0.013	0.003	0.000	0.026	0.497	2.333	0.524	0.000	0.004	0.008	0.720	0.224	0.346	0.009	0.030
23	65.3	0.019	0.001	0.007	0.000	0.055	1.615	1.007	0.223	0.005	0.001	0.000	0.828	0.101	0.207	0.000	0.013
24	70.0	0.028	0.000	0.008	0.000	0.087	2.776	0.531	0.166	0.000	0.001	0.000	0.913	0.055	0.178	0.000	0.012
25	66.2	0.014	0.003	0.007	0.000	0.058	1.428	1.898	0.525	0.000	0.008	0.014	0.907	0.255	0.368	0.000	0.034
26	42.7	0.036	0.017	0.011	0.000	0.063	0.991	2.495	0.457	0.000	0.000	0.000	0.529	0.030	0.094	0.000	0.000

Host: Ocean Song

Del: delphinidin,

Cya: cyanidin,

Pet: petunidin,

Pel: pelargonidin,

Mal: malvidin

M: myricetin,

Q: quercetin,

K: kaempferol

Tri: tricetin,

Lut: luteolin,

Api: apigenin,

Vic2: vicenin-2,

VX: vitexin,

IVX: isovitexin,

Ori: orientin,

Iori: isoorientin

Mal(%): Proportion of malvidin in total anthocyanidins

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This plasmid constitutively expresses the *Campanula* F3',5'H gene, *torenia* MT gene, *licorice* F2H gene, rice codon usage-modified CGT gene and *Lotus japonicus* FDH gene in plants.

The constructed plasmid pSPB6486 was introduced into the blue rose variety "Ocean Song", and a total of 27 transformants were obtained. Upon pigment analysis, malvidin storage was confirmed in 26 transformants, with a maximum malvidin content of 74.5% (average: 57.0%). The flavone C-glycosides isovitexin (apigenin 6-C-glucoside), vitexin (apigenin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside), orientin (luteolin 8-C-glucoside) and vicenin-2 (apigenin 6,8-di-C-glucoside) were also identified and quantified. Flavone C-glycosides were detected in all of the transformants in which malvidins were detected, with the flavone di-C-glycoside vicenin-2 as the major detected component. The amounts of production were greater than in the

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Example 3: Transfer of *Campanula*-Derived F3',5'H Gene, *Torenia*-Derived MT Gene, *Licorice*-Derived F2H Gene, *Buckwheat*-Derived Codon Usage-Modified CGT Gene and *Lotus japonicus*-Derived FDH Gene into Rose Variety "Ocean Song"

Plasmid pSPB7473 has pBINPLUS as the basic backbone, and contains the following four expression cassettes.

(1) El₂35S promoter, *Campanula*-derived F3',5'H full-length cDNA (SEQ ID NO: 1) and D8 terminator

(2) El₂35S promoter, *torenia*-derived MT full-length cDNA (SEQ ID NO: 3) (*Arabidopsis thaliana* HSPRO gene-derived 5'-UTR (SEQ ID NO: 13) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

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(3) El₂35S promoter, licorice-derived F2H full-length cDNA (SEQ ID NO: 5) and *perilla*-derived AT terminator

(4) El₂35S promoter, buckwheat-derived codon usage-modified CGT full-length cDNA (SEQ ID NO: 11) (*Arabidopsis thaliana* HSPRO gene-derived 5'-UTR (SEQ ID NO: 13) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

(5) El₂35S promoter, *Lotus japonicus*-derived FDH full-length cDNA (SEQ ID NO: 9) and *Arabidopsis thaliana*-derived HSP terminator

This plasmid constitutively expresses the *Campanula* F3',5'H gene, *torenia* MT gene, licorice F2H gene, buckwheat codon usage-modified CGT gene and *Lotus japonicus* FDH gene in plants.

The constructed plasmid pSPB7473 was introduced into the blue rose variety "Ocean Song", and a total of 35 transformants were obtained. Upon pigment analysis, malvidin storage was confirmed in 21 transformants, with a

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maximum malvidin content of 20.8% (average: 9.1%). The flavone C-glycosides isovitexin (apigenin 6-C-glucoside), vitexin (apigenin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside), orientin (luteolin 8-C-glucoside) and vicenin-2 (apigenin 6,8-di-C-glucoside) were also identified and quantified. Flavone C-glycosides were detected in all of the transformants in which malvidins were detected, with the flavone 6-C-glycoside isovitexin as the major detected component. The amounts of production were greater in the order isovitexin >vitexin >vicenin-2>isoorientin >orientin, and the maximum total amount was 7.418 mg per 1 g of fresh petal weight, indicating a very high content. In addition, in comparison to the OS/6486 line described in Example 2, the total amount of flavone C-glycosides was a high content of 3 mg or greater per 1 g of fresh petal weight in most of the transformants, which was at least about 100 times compared to delphinidins.

The measured values for representative transformants are shown in Table 3 below.

TABLE 3

Plant	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				Mal	Total flavone C- glyco- sides (mg/g fresh
No.	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori	(%)	weight)
Host	0.000	0.024	0.000	0.000	0.000	0.000	2.883	0.586	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	—	—
1	0.053	0.001	0.005	0.000	0.003	1.261	0.220	0.044	0.029	0.000	0.000	0.922	1.500	1.927	0.083	0.169	5.4	4.601
2	0.065	0.000	0.007	0.000	0.003	2.520	0.130	0.051	0.042	0.000	0.000	1.773	2.140	2.893	0.141	0.288	4.2	7.235
3	0.086	0.004	0.009	0.000	0.005	1.459	0.419	0.093	0.000	0.000	0.000	1.160	1.777	2.203	0.082	0.165	4.8	5.387
4	0.054	0.000	0.008	0.000	0.009	2.885	0.168	0.032	0.000	0.000	0.000	0.041	0.000	0.000	0.000	0.000	13.0	0.041
5	0.063	0.000	0.009	0.000	0.007	2.903	0.186	0.055	0.041	0.000	0.000	1.787	2.255	2.962	0.138	0.276	8.7	7.418
6	0.009	0.004	0.002	0.000	0.002	1.096	2.107	0.345	0.026	0.000	0.026	1.093	1.774	2.711	0.095	0.177	9.6	5.849
7	0.012	0.008	0.002	0.000	0.001	0.683	1.642	0.173	0.025	0.000	0.015	0.800	1.192	1.648	0.063	0.138	6.2	3.840
8	0.011	0.001	0.003	0.000	0.002	1.737	0.984	0.164	0.029	0.000	0.010	0.990	1.582	2.389	0.100	0.195	12.5	5.256
9	0.026	0.016	0.002	0.000	0.001	0.723	1.928	0.181	0.000	0.000	0.012	0.796	1.080	1.320	0.000	0.114	3.0	3.310
10	0.022	0.008	0.002	0.000	0.002	1.017	1.325	0.309	0.000	0.000	0.000	1.394	1.442	2.123	0.070	0.136	5.5	5.165
11	0.040	0.000	0.004	0.000	0.002	1.706	0.177	0.060	0.019	0.000	0.000	0.851	1.086	1.541	0.068	0.129	4.1	3.674
12	0.035	0.000	0.003	0.000	0.002	1.898	0.219	0.069	0.023	0.000	0.000	1.000	1.328	1.892	0.082	0.165	4.6	4.466
13	0.031	0.000	0.004	0.000	0.003	0.888	0.041	0.012	0.027	0.004	0.000	1.120	1.482	1.939	0.091	0.194	7.8	4.826
14	0.040	0.000	0.005	0.000	0.005	0.980	0.048	0.017	0.024	0.000	0.000	1.153	1.387	1.757	0.087	0.183	9.1	4.568
15	0.027	0.000	0.007	0.000	0.009	1.273	0.108	0.030	0.000	0.000	0.000	0.533	1.386	1.402	0.059	0.122	20.8	3.501
16	0.020	0.000	0.005	0.000	0.006	1.567	0.129	0.045	0.025	0.000	0.008	0.610	1.523	1.819	0.084	0.165	18.9	4.200
17	0.018	0.051	0.003	0.000	0.004	0.188	1.761	0.118	0.000	0.000	0.000	0.705	0.409	0.632	0.016	0.043	5.1	1.806
18	0.014	0.035	0.002	0.000	0.004	0.242	2.042	0.265	0.010	0.000	0.000	1.008	0.682	1.022	0.034	0.066	7.7	2.813
19	0.021	0.028	0.005	0.000	0.014	0.339	1.434	0.167	0.012	0.000	0.000	0.935	0.919	1.241	0.037	0.081	19.7	3.214
20	0.014	0.033	0.002	0.000	0.006	0.236	1.697	0.204	0.000	0.000	0.000	1.155	0.839	1.265	0.036	0.086	10.0	3.381
21	0.045	0.001	0.006	0.000	0.006	1.641	0.302	0.087	0.018	0.000	0.000	0.987	1.066	1.296	0.051	0.099	10.9	3.499
Average																	9.1	4.193
S.D.																	5.3	1.663

Host: Ocean Song

Del: delphinidin,

Cya: cyanidin,

Pet: petunidin,

Pel: pelargonidin,

Mal: malvidin

M: myricetin,

Q: quercetin,

K: kaempferol

Tri: tricetin,

Lut: luteolin,

Api: apigenin,

Vic2: vicenin-2,

VX: vitexin,

IVX: isovitexin,

Ori: orientin,

Iori: isoorientin

Mal(%): Proportion of malvidin in total anthocyanidins

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Example 4: Transfer of *Campanula*-Derived F3',5'H Gene, *Torenia*-Derived MT Gene, Licorice-Derived F2H Gene, Buckwheat-Derived CGT Gene and *Lotus japonicus*-Derived FDH Gene into Rose Variety "Ocean Song"

Plasmid pSPB7472 has pBINPLUS as the basic backbone, and contains the following four expression cassettes.

(1) El₂35S promoter, *Campanula*-derived F3',5'H full-length cDNA (SEQ ID NO: 1) and D8 terminator

(2) El₂35S promoter, *torenia*-derived MT full-length cDNA (SEQ ID NO: 3) (*Arabidopsis thaliana* HSPRO gene-derived 5'-UTR (SEQ ID NO: 13) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

(3) El₂35S promoter, licorice-derived F2H full-length cDNA (SEQ ID NO: 5) and *perilla*-derived AT terminator

(4) El₂35S promoter, buckwheat-derived CGT full-length cDNA (SEQ ID NO: 14) (*Arabidopsis thaliana* HSPRO

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transformants were obtained. The flavone C-glycosides isovitexin (apigenin 6-C-glucoside), vitexin (apigenin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside), orientin (luteolin 8-C-glucoside) and vicenin-2 (apigenin 6,8-di-C-glucoside), and the anthocyanidins delphinidin, cyanidin, petunidin, pelargonidin and malvidin, were then identified and quantified. As a result, accumulation of flavone C-glycosides and malvidin was confirmed in 11 transformants. The average content of flavone C-glycosides was 3.75 mg per 1 g of fresh petal weight, with the flavone 6-C-glycoside isovitexin detected as the main component. The malvidin content was a maximum of 15.6% (average: 8.8%).

Thus, the average content of flavone C-glycosides per 1 g of fresh petal weight was higher in the OS/7473 line described in Example 3, at 4.19 mg. In other words, it is possible to produce flavone C-glycosides more efficiently with the codon-modified CGT gene than with the original buckwheat-derived CGT gene.

The measured values for representative transformants are shown in Table 4 below.

TABLE 4

Plant No.	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				Total flavone C-glycosides (mg/g fresh weight)
	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori	(%)
Host	0.000	0.024	0.000	0.000	0.000	0.000	2.883	0.586	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	—
1	0.120	0.000	0.027	0.000	0.027	2.784	0.211	0.029	0.000	0.000	0.000	0.061	0.000	0.000	0.000	0.000	15.6
2	0.041	0.000	0.007	0.000	0.003	2.485	0.369	0.094	0.036	0.000	0.000	1.010	1.873	2.654	0.107	0.221	6.8
3	0.100	0.002	0.008	0.000	0.002	1.632	0.316	0.055	0.000	0.000	0.000	1.244	2.093	2.482	0.103	0.204	1.9
4	0.025	0.000	0.002	0.000	0.001	1.916	0.221	0.060	0.026	0.000	0.000	0.872	1.545	2.228	0.104	0.200	4.1
5	0.031	0.000	0.005	0.000	0.003	2.125	0.298	0.076	0.021	0.000	0.000	0.847	1.400	1.937	0.075	0.156	6.7
6	0.039	0.000	0.008	0.000	0.005	1.926	0.251	0.072	0.000	0.000	0.000	0.717	1.292	1.731	0.070	0.143	8.9
7	0.043	0.000	0.004	0.000	0.003	1.117	0.126	0.031	0.024	0.000	0.000	1.304	1.365	1.781	0.082	0.177	5.3
8	0.053	0.002	0.010	0.000	0.010	0.968	0.143	0.023	0.000	0.000	0.000	0.982	1.064	1.495	0.062	0.129	13.2
9	0.016	0.000	0.005	0.000	0.004	1.759	0.175	0.083	0.000	0.000	0.000	0.661	0.941	2.106	0.100	0.174	14.9
10	0.014	0.000	0.004	0.000	0.003	1.618	0.148	0.088	0.000	0.000	0.000	0.651	0.820	1.741	0.000	0.153	13.4
11	0.011	0.000	0.002	0.000	0.001	0.292	0.000	0.000	0.332	0.433	0.082	0.095	0.000	0.000	0.000	0.000	6.3
Average																	8.8
S.D.																	4.7

Host: Ocean Song

Del: delphinidin,

Cya: cyanidin,

Pet: petunidin,

Pel: pelargonidin,

Mal: malvidin

M: myricetin,

Q: quercetin,

K: kaempferol

Tri: tricetin,

Lut: luteolin,

Api: apigenin,

Vic2: vicenin-2,

VX: vitexin,

IVX: isovitexin,

Ori: orientin,

Iori: isoorientin

Mal(%): Proportion of malvidin in total anthocyanidins

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gene-derived 5'-UTR (SEQ ID NO: 13)) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

(5) El₂35S promoter, *Lotus japonicus*-derived FDH full-length cDNA (SEQ ID NO: 9) and *Arabidopsis thaliana*-derived HSP terminator

This plasmid constitutively expresses the *Campanula* F3',5'H gene, *torenia* MT gene, licorice F2H gene, buckwheat CGT gene and *Lotus japonicus* FDH gene in plants.

The constructed plasmid pSPB7472 was introduced into the blue rose variety "Ocean Song", and a total of 33

Example 5: Transfer of *Campanula*-Derived F3',5'H Gene, *Torenia*-Derived MT Gene, Licorice-Derived F2H Gene, Buckwheat-Derived CGT Gene and *Lotus japonicus*-Derived FDH Gene into Rose Variety "Ocean Song"

Plasmid pSPB7808 has pBINPLUS as the basic backbone, and contains the following four expression cassettes.

(1) El₂35S promoter, *Campanula*-derived F3',5'H full-length cDNA (SEQ ID NO: 1) and D8 terminator

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(2) El₂35S promoter, torenia-derived MT full-length cDNA (SEQ ID NO: 3) and *Arabidopsis thaliana*-derived HSP terminator

(3) El₂35S promoter, licorice-derived F2H full-length cDNA (SEQ ID NO: 5) and *perilla*-derived AT terminator

(4) El₂35S promoter, buckwheat-derived CGT full-length cDNA (SEQ ID NO: 14) (*Arabidopsis thaliana* alcohol dehydrogenase (ADH) gene-derived 5'-UTR (SEQ ID NO: 15) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

(5) El₂35S promoter, *Lotus japonicus*-derived FDH full-length cDNA (SEQ ID NO: 9) and *Arabidopsis thaliana*-derived HSP terminator

This plasmid constitutively expresses the *Campanula* F3',5'H gene, torenia MT gene, licorice F2H gene, buckwheat CGT gene and *Lotus japonicus* FDH gene in plants.

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The constructed plasmid pSPB7808 was introduced into the blue rose variety "Ocean Song", and a total of 65 transformants were obtained. The flavone C-glycosides isovitexin (apigenin 6-C-glucoside), vitexin (apigenin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside), orientin (luteolin 8-C-glucoside) and vicenin-2 (apigenin 6,8-di-C-glucoside), and the anthocyanidins delphinidin, cyanidin, petunidin, pelargonidin and malvidin, were then identified and quantified. As a result, accumulation of flavone C-glycosides and malvidin was confirmed in 32 transformants. The mean content of flavone C-glycosides was 3.00 mg per 1 g of fresh petal weight, with the flavone 6-C-glucoside isovitexin detected as the main component. The malvidin content was a maximum of 69.2% (average: 43.9%).

The measured values for representative transformants are shown in Table 5 below.

TABLE 5

Plant	Flavonoid compounds (mg/g fresh weight)																Total flavone C-glycosides (mg/g fresh weight)	
	Anthocyanidin				Flavonol				Flavone				Flavone C-glycoside				Mal	fresh
	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori		
No.	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori	(%)	weight)
Host	0.000	0.024	0.000	0.000	0.000	0.000	2.883	0.586	0.000	0.000	0.000	0.000	0.000	0.000	0.000	—	—	
1	0.052	0.000	0.004	0.000	0.002	0.755	0.000	0.000	0.074	0.174	0.086	0.329	1.123	2.417	0.145	0.285	3.0	4.299
2	0.026	0.002	0.020	0.000	0.068	1.448	0.321	0.206	0.000	0.000	0.000	0.560	0.587	1.380	0.000	0.056	57.1	2.583
3	0.021	0.000	0.010	0.000	0.026	1.569	0.271	0.252	0.000	0.000	0.000	0.898	0.878	1.945	0.051	0.108	46.7	3.880
4	0.022	0.000	0.009	0.000	0.023	1.549	0.340	0.265	0.000	0.000	0.000	0.867	0.874	1.792	0.044	0.090	42.5	3.667
5	0.006	0.000	0.004	0.000	0.005	0.775	0.000	0.000	0.027	0.146	0.120	0.238	0.431	1.005	0.063	0.115	34.4	1.851
6	0.006	0.000	0.003	0.000	0.002	0.747	0.000	0.000	0.025	0.110	0.083	0.237	0.389	0.907	0.065	0.116	18.8	1.715
7	0.007	0.000	0.003	0.000	0.007	1.014	0.064	0.028	0.029	0.148	0.100	0.410	0.596	1.338	0.099	0.176	41.0	2.619
8	0.007	0.000	0.003	0.000	0.006	1.134	0.080	0.025	0.034	0.183	0.127	0.354	0.638	1.423	0.101	0.179	36.9	2.696
9	0.009	0.000	0.004	0.000	0.011	1.713	0.303	0.127	0.000	0.000	0.000	0.617	0.682	1.498	0.052	0.101	46.6	2.949
10	0.006	0.000	0.005	0.000	0.016	1.359	0.292	0.128	0.000	0.000	0.000	0.545	0.581	1.387	0.000	0.068	60.2	2.582
11	0.006	0.000	0.003	0.000	0.003	0.809	0.054	0.020	0.025	0.124	0.101	0.184	0.501	1.138	0.074	0.130	26.2	2.027
12	0.004	0.000	0.003	0.000	0.005	0.692	0.065	0.031	0.020	0.122	0.107	0.186	0.484	1.109	0.065	0.115	42.3	1.959
13	0.006	0.010	0.003	0.000	0.015	0.403	1.049	0.445	0.000	0.000	0.000	0.292	0.144	0.260	0.000	0.000	36.1	0.695
14	0.005	0.002	0.002	0.000	0.026	0.594	0.363	0.193	0.000	0.000	0.000	0.403	0.216	0.545	0.000	0.000	69.2	1.165
15	0.008	0.002	0.003	0.000	0.027	1.171	1.128	0.196	0.000	0.000	0.000	1.749	1.080	1.891	0.000	0.101	61.8	4.821
16	0.006	0.000	0.004	0.000	0.005	0.880	0.064	0.022	0.030	0.162	0.142	0.286	0.633	1.469	0.088	0.141	30.5	2.617
17	0.015	0.000	0.009	0.000	0.016	2.291	0.190	0.064	0.065	0.322	0.220	0.440	1.307	3.073	0.205	0.381	40.4	5.406
18	0.008	0.000	0.004	0.000	0.003	1.684	0.157	0.031	0.045	0.155	0.168	0.543	1.149	2.038	0.131	0.238	22.3	4.099
19	0.005	0.000	0.003	0.000	0.007	0.712	0.074	0.026	0.000	0.000	0.000	0.558	0.346	0.816	0.000	0.067	48.0	1.788
20	0.005	0.000	0.003	0.000	0.007	0.774	0.080	0.027	0.000	0.000	0.000	0.479	0.356	0.848	0.000	0.068	47.5	1.751
21	0.005	0.000	0.003	0.000	0.009	0.614	0.054	0.000	0.000	0.000	0.000	0.427	0.294	0.699	0.000	0.056	51.1	1.477
22	0.005	0.000	0.005	0.000	0.013	1.117	0.179	0.075	0.000	0.000	0.000	0.481	0.414	1.023	0.000	0.063	57.5	1.981
23	0.007	0.000	0.006	0.000	0.019	1.374	0.257	0.102	0.000	0.000	0.000	0.551	0.478	1.178	0.000	0.068	59.9	2.275
24	0.006	0.000	0.004	0.000	0.014	1.181	0.228	0.082	0.000	0.000	0.000	0.433	0.403	0.984	0.000	0.057	58.9	1.877
25	0.010	0.000	0.005	0.000	0.011	0.641	0.096	0.019	0.000	0.000	0.000	1.121	0.642	1.207	0.000	0.122	42.6	3.092
26	0.008	0.000	0.005	0.000	0.013	1.270	0.149	0.066	0.000	0.000	0.000	1.751	1.287	1.468	0.000	0.143	50.8	4.649
27	0.020	0.000	0.009	0.000	0.040	0.471	0.095	0.000	0.000	0.000	0.000	1.635	1.750	2.498	0.000	0.176	57.9	6.059
28	0.021	0.000	0.010	0.000	0.027	0.947	0.259	0.060	0.000	0.000	0.000	1.081	0.848	1.753	0.000	0.128	46.7	3.810
29	0.018	0.002	0.009	0.000	0.009	0.589	0.079	0.017	0.000	0.065	0.060	0.567	0.640	1.344	0.000	0.155	23.2	2.706
30	0.010	0.000	0.005	0.000	0.011	1.521	0.330	0.148	0.020	0.014	0.032	1.055	0.877	1.639	0.000	0.124	41.9	3.694
31	0.008	0.000	0.005	0.000	0.014	2.170	0.595	0.163	0.000	0.000	0.000	1.406	1.255	2.150	0.000	0.157	51.5	4.969
32	0.006	0.000	0.003	0.000	0.009	1.998	0.519	0.142	0.000	0.000	0.000	1.156	1.098	1.854	0.068	0.143	49.6	4.318

TABLE 5-continued

Plant	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				Total flavone C- glycosides (mg/g fresh weight)	
	No.	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori		Iori
Average																		43.9
S.D.																		3.002
																		14.4
																		1.329

Host: Ocean Song

Del: delphinidin,

Cya: cyanidin,

Pet: petunidin,

Pel: pelargonidin,

Mal: malvidin

M: myricetin,

Q: quercetin,

K: kaempferol

Tri: tricetin,

Lut: luteolin,

Api: apigenin,

Vic2: vicenin-2,

VX: vitexin,

IVX: isovitexin,

Ori: orientin,

Iori: isoorientin

Mal(%): Proportion of malvidin in total anthocyanidins

Example 6: Transfer of *Campanula*-Derived F3',5'H Gene, *Torenia*-Derived MT Gene, *Licorice*-Derived F2H Gene, *Buckwheat*-Derived Codon Usage-Modified CGT Gene and *Lotus japonicus*-Derived FDH Gene into Rose Variety "Ocean Song"

Plasmid pSPB7809 has pBINPLUS as the basic backbone, and contains the following four expression cassettes.

(1) El₂35S promoter, *Campanula*-derived F3',5'H full-length cDNA (SEQ ID NO: 1) and D8 terminator

(2) El₂35S promoter, *torenia*-derived MT full-length cDNA (SEQ ID NO: 3) and *Arabidopsis thaliana*-derived HSP terminator

(3) El₂35S promoter, *licorice*-derived F2H full-length cDNA (SEQ ID NO: 5) and *perilla*-derived AT terminator

(4) El₂35S promoter, *buckwheat*-derived codon usage-modified CGT full-length cDNA (SEQ ID NO: 11) (*Arabidopsis thaliana* alcohol dehydrogenase (ADH) gene-derived 5'-UTR (SEQ ID NO: 15) added to the 5'-position end) and *Arabidopsis thaliana*-derived HSP terminator

(5) El₂35S promoter, *Lotus japonicus*-derived FDH full-length cDNA (SEQ ID NO: 9) and *Arabidopsis thaliana*-derived HSP terminator

This plasmid constitutively expresses the *Campanula* F3',5'H gene, *torenia* MT gene, *licorice* F2H gene, *buckwheat* codon usage-modified CGT gene and *Lotus japonicus* FDH gene in plants.

The constructed plasmid pSPB7809 was introduced into the blue rose variety "Ocean Song", and a total of 143 transformants were obtained. The flavone C-glycosides isovitexin (apigenin 6-C-glucoside), vitexin (apigenin 8-C-glucoside), isoorientin (luteolin 6-C-glucoside), orientin (luteolin 8-C-glucoside) and vicenin-2 (apigenin 6,8-di-C-glucoside), and the anthocyanidins delphinidin, cyanidin, petunidin, pelargonidin and malvidin, were then identified and quantified. As a result, accumulation of flavone C-glycosides was confirmed in 58 transformants. The mean content of flavone C-glycosides was 3.24 mg per 1 g of fresh petal weight, with the flavone 6-C-glycoside isovitexin detected as the main component. The malvidin content was a maximum of 80.3% (average: 46.6%).

Thus, the average content of flavone C-glycosides per 1 g of fresh petal weight was higher in the OS/7809 line described in the Examples. Similar to the results obtained in Examples 3 and 4, it was shown to be possible to produce flavone C-glycosides more efficiently with the codon-modified CGT gene than with the original buckwheat-derived CGT gene.

The measured values for representative transformants are shown in Table 6 below.

TABLE 6

Plant	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				Mal	Total flavone C- glycosides (mg/g fresh weight)	
	No.	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori			Iori
Host	0.000	0.024	0.000	0.000	0.000	0.000	0.576	2.883	0.586	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	—	—
1	0.019	0.000	0.003	0.000	0.002	0.033	0.865	0.068	0.020	0.083	0.137	0.071	0.445	1.009	2.121	0.136	0.280	7.0	3.991
2	0.011	0.000	0.005	0.000	0.033	0.033	0.865	0.091	0.042	0.000	0.000	0.000	1.133	0.673	1.375	0.047	0.106	67.1	3.335
3	0.011	0.000	0.005	0.000	0.028	0.018	1.183	0.119	0.068	0.000	0.000	0.000	0.954	0.602	1.284	0.053	0.096	63.9	2.988
4	0.009	0.000	0.005	0.000	0.017	0.017	1.332	0.177	0.093	0.000	0.000	0.000	0.942	0.498	1.150	0.041	0.089	54.4	2.720
5	0.015	0.000	0.007	0.000	0.018	0.018	1.521	0.209	0.107	0.000	0.000	0.000	1.223	0.748	1.707	0.062	0.139	45.8	3.879

TABLE 6-continued

Plant	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				Total flavone C- glycosides (mg/g fresh
	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori	Iori	(%) weight)
6	0.013	0.000	0.007	0.000	0.017	1.007	0.131	0.064	0.000	0.000	0.000	0.844	0.523	1.181	0.047	0.101	46.1 2.694
7	0.017	0.000	0.006	0.000	0.018	1.852	0.363	0.200	0.000	0.000	0.000	0.759	0.711	1.482	0.042	0.091	44.5 3.083
8	0.009	0.000	0.006	0.000	0.018	1.395	0.253	0.130	0.000	0.000	0.000	0.819	0.819	1.790	0.052	0.113	53.2 3.593
9	0.007	0.000	0.002	0.000	0.006	0.694	0.000	0.000	0.057	0.254	0.122	0.381	0.470	1.075	0.105	0.187	38.8 2.219
10	0.011	0.000	0.003	0.000	0.003	0.711	0.000	0.000	0.062	0.222	0.099	0.365	0.533	1.243	0.117	0.214	18.9 2.472
11	0.010	0.000	0.005	0.000	0.007	0.772	0.000	0.000	0.023	0.104	0.090	0.481	0.451	1.070	0.062	0.122	32.1 2.186
12	0.009	0.000	0.005	0.000	0.012	1.093	0.121	0.066	0.000	0.000	0.000	0.618	0.508	1.105	0.051	0.078	47.4 2.360
13	0.006	0.000	0.004	0.000	0.008	0.858	0.070	0.027	0.000	0.000	0.000	1.033	0.874	1.896	0.071	0.163	44.1 4.038
14	0.035	0.003	0.005	0.000	0.003	1.213	0.803	0.673	0.000	0.000	0.000	0.599	0.064	0.263	0.000	0.000	5.7 0.926
15	0.011	0.000	0.003	0.000	0.002	0.921	0.000	0.000	0.066	0.243	0.117	0.387	0.685	1.543	0.129	0.218	15.2 2.962
16	0.005	0.000	0.003	0.000	0.011	0.540	0.087	0.075	0.000	0.000	0.000	0.841	0.569	1.286	0.000	0.085	58.1 2.781
17	0.004	0.000	0.002	0.000	0.007	0.878	0.000	0.025	0.044	0.202	0.135	0.451	0.619	1.438	0.114	0.198	53.0 2.820
18	0.004	0.000	0.003	0.000	0.019	1.616	0.106	0.000	0.000	0.000	0.000	0.722	0.515	1.201	0.043	0.115	73.7 2.595
19	0.007	0.000	0.005	0.000	0.043	2.309	0.235	0.097	0.000	0.000	0.000	0.655	0.625	1.409	0.000	0.097	77.2 2.786
20	0.016	0.000	0.005	0.000	0.009	0.771	0.079	0.035	0.056	0.155	0.119	0.532	0.812	1.796	0.100	0.193	30.2 3.433
21	0.016	0.000	0.008	0.000	0.011	0.788	0.061	0.027	0.062	0.123	0.081	0.749	1.191	2.623	0.126	0.262	32.0 4.951
22	0.010	0.000	0.005	0.000	0.007	0.884	0.055	0.000	0.030	0.116	0.085	0.514	0.679	1.530	0.085	0.156	32.8 2.964
23	0.006	0.000	0.003	0.000	0.031	0.947	0.171	0.082	0.000	0.000	0.000	1.412	0.844	2.030	0.058	0.113	76.8 4.457
24	0.011	0.000	0.004	0.000	0.028	1.048	0.173	0.091	0.000	0.000	0.000	0.959	0.608	1.396	0.048	0.102	65.0 3.113
25	0.015	0.000	0.003	0.000	0.005	1.028	0.000	0.000	0.050	0.134	0.071	0.336	0.997	1.946	0.139	0.245	21.8 3.664
26	0.006	0.000	0.004	0.000	0.010	0.443	0.000	0.000	0.165	0.397	0.114	0.330	0.000	0.418	0.045	0.110	49.1 0.903
27	0.005	0.000	0.003	0.000	0.008	0.513	0.000	0.000	0.124	0.604	0.259	0.267	0.157	0.397	0.040	0.074	47.5 0.935
28	0.007	0.000	0.004	0.000	0.012	0.363	0.000	0.000	0.092	0.327	0.104	0.207	0.103	0.291	0.000	0.066	51.5 0.667
29	0.015	0.000	0.011	0.000	0.026	1.208	0.161	0.059	0.000	0.000	0.000	1.856	0.926	2.140	0.079	0.229	49.6 5.229
30	0.013	0.000	0.009	0.000	0.020	1.138	0.097	0.039	0.000	0.000	0.000	1.499	0.819	1.941	0.070	0.198	47.5 4.526
31	0.008	0.000	0.004	0.000	0.007	1.354	0.112	0.036	0.019	0.093	0.090	0.395	0.752	1.661	0.089	0.154	35.0 3.051
32	0.005	0.000	0.003	0.000	0.009	1.819	0.287	0.181	0.000	0.000	0.000	1.436	1.025	2.410	0.064	0.194	52.6 5.128
33	0.007	0.000	0.005	0.000	0.012	1.233	0.148	0.072	0.000	0.000	0.000	0.975	0.604	1.441	0.055	0.109	51.4 3.185
34	0.005	0.000	0.003	0.000	0.011	2.099	0.333	0.154	0.000	0.000	0.000	1.263	0.733	1.737	0.056	0.116	57.4 3.905
35	0.014	0.000	0.009	0.000	0.024	1.248	0.283	0.131	0.000	0.000	0.000	1.183	0.760	1.884	0.043	0.146	51.2 4.016
36	0.014	0.000	0.003	0.000	0.002	0.375	0.034	0.000	0.033	0.092	0.049	0.350	0.479	1.099	0.079	0.134	10.6 2.142
37	0.010	0.000	0.002	0.000	0.001	0.705	0.077	0.021	0.031	0.068	0.056	0.426	0.620	1.405	0.075	0.135	9.3 2.660
38	0.003	0.000	0.002	0.000	0.017	1.087	0.153	0.066	0.000	0.000	0.000	1.481	0.744	1.612	0.050	0.109	80.3 3.995
39	0.001	0.000	0.000	0.000	0.004	0.988	0.469	0.204	0.000	0.000	0.000	0.805	0.661	1.547	0.043	0.073	72.8 3.128
40	0.003	0.003	0.002	0.000	0.011	0.992	1.325	0.414	0.000	0.000	0.000	1.271	0.853	1.871	0.047	0.097	51.4 4.140
41	0.006	0.000	0.002	0.000	0.017	1.430	0.621	0.293	0.000	0.000	0.000	0.812	0.741	1.633	0.053	0.087	68.3 3.325
42	0.004	0.000	0.002	0.000	0.014	1.223	0.356	0.195	0.000	0.000	0.000	0.981	0.861	1.781	0.000	0.076	70.9 3.699
43	0.008	0.000	0.004	0.000	0.023	1.211	0.512	0.350	0.000	0.000	0.000	1.026	0.935	2.040	0.000	0.081	66.6 4.081
44	0.009	0.000	0.005	0.000	0.008	1.333	0.147	0.037	0.053	0.203	0.088	0.187	0.337	0.796	0.079	0.139	36.4 1.537
45	0.020	0.001	0.011	0.000	0.016	1.182	0.000	0.022	0.093	0.348	0.140	0.255	0.349	0.777	0.084	0.163	32.9 1.629
46	0.016	0.000	0.009	0.000	0.016	1.232	0.125	0.038	0.074	0.289	0.114	0.160	0.390	0.844	0.088	0.159	39.2 1.641
47	0.004	0.000	0.003	0.000	0.005	0.824	0.057	0.000	0.033	0.090	0.088	0.555	1.502	3.439	0.152	0.276	42.9 5.924
48	0.007	0.000	0.005	0.000	0.016	1.366	0.349	0.123	0.000	0.000	0.000	0.672	0.682	1.510	0.000	0.086	56.4 2.951
49	0.009	0.000	0.005	0.000	0.017	1.407	0.382	0.138	0.000	0.000	0.000	0.594	0.654	1.456	0.039	0.084	55.1 2.828
50	0.006	0.000	0.003	0.000	0.033	2.879	0.358	0.065	0.000	0.000	0.000	1.522	0.782	1.397	0.000	0.135	77.9 3.836
51	0.018	0.000	0.008	0.000	0.020	0.765	0.147	0.027	0.000	0.000	0.000	1.224	0.584	1.174	0.000	0.108	43.7 3.090
52	0.021	0.000	0.010	0.000	0.027	1.359	0.357	0.183	0.000	0.000	0.000	2.105	0.689	1.831	0.000	0.109	46.5 4.734
53	0.009	0.000	0.006	0.000	0.016	1.485	0.248	0.073	0.000	0.000	0.000	1.808	0.630	1.256	0.000	0.113	52.7 3.807
54	0.005	0.000	0.003	0.000	0.018	1.235	0.610	0.220	0.000	0.000	0.000	1.640	1.201	1.893	0.000	0.084	68.3 4.818
55	0.007	0.000	0.004	0.000	0.006	1.082	0.086	0.019	0.034	0.076	0.096	0.569	1.517	2.754	0.133	0.262	34.9 5.234
56	0.008	0.000	0.003	0.000	0.002	0.998	0.000	0.000	0.072	0.253	0.166	0.457	0.667	1.062	0.092	0.183	18.7 2.461
57	0.017	0.000	0.005	0.000	0.006	1.031	0.000	0.000	0.078	0.120	0.064	0.543	1.090	1.621	0.151	0.286	21.9 3.691
58	0.010	0.000	0.006	0.000	0.014	1.598	0.292	0.111	0.000	0.000	0.000	1.511	0.931	1.176	0.000	0.098	46.7 3.717

TABLE 6-continued

																		Total flavone C-glycosides (mg/g fresh Mal
Plant	Anthocyanidin (mg/g fresh weight)				Flavonol (mg/g fresh weight)				Flavone (mg/g fresh weight)				Flavone C-glycoside (mg/g fresh weight)				weight)	
	No.	Del	Cya	Pet	Pel	Mal	M	Q	K	Tri	Lut	Api	Vic2	VX	IVX	Ori		Iori
Average																		46.6
S.D.																		3.235
																		18.9
																		1.143

Host: Ocean Song

Del: delphinidin,

Cya: cyanidin,

Pet: petunidin,

Pel: pelargonidin,

Mal: malvidin

M: myricetin,

Q: quercetin,

K: kaempferol

Tri: tricetin,

Lut: luteolin,

Api: apigenin,

Vic2: vicenin-2,

VX: vitexin,

IVX: isovitexin,

Ori: orientin,

Iori: isoorientin

Mal(%): Proportion of malvidin in total anthocyanidins

Example 7: Evaluation of Flower Color of Flavone C-Glycoside-Containing Roses

The transformants created in Examples 2 and 3 (with “Ocean Song” rose variety as host) were measured to determine the color shades of the respective petals using a CM-2022 spectrophotometer (Minolta) with a 10° visual field and a D65 light source, and analyzing with Spectra-Magic™ color management software (Minolta).

In comparing the average values of the hue angles, no differences in petal hue angle were found between the roses with buckwheat-derived (Example 3) and roses with rice-derived (Example 2) CGT genes. On the individual level, however, more transformants exhibited a hue angle of 315° or smaller among the roses with the transferred buckwheat-derived CGT gene, with one having been altered to the bluest color yet obtained, having a value of 294.5°. These results confirmed that using the buckwheat-derived CGT gene significantly increased the amount of flavone C-glycoside production, and especially the amount of mono-C-glycoside production, altering the petals to a blue color shade by their copresence with anthocyanins.

The results are shown in Table 7.

TABLE 7

	Gene and flavonoid composition	Hue angle (hue)
Host	Ocean Song Stores cyanidin as main pigment, contains absolutely no flavone C-glycoside	Average: 362.57°
(1) <i>Campanula</i> F3',5'H	Has high storage of delphinidin as main pigment, contains absolutely no flavone C-glycoside	Average: 337.33°, bluest point: 333.15°
(2) <i>Campanula</i> F3',5'H + torenia MT + licorice F2H + rice CGT (codon usage modified) + <i>Lotus japonicus</i> FDH	Has high storage of malvidin as main pigment, contains flavone C-glycoside	Average: 320.16°, bluest point: 318.89°
(3) <i>Campanula</i> F3',5'H + torenia MT + licorice F2H + buckwheat CGT (codon usage modified) + <i>Lotus japonicus</i> FDH	Has high storage of delphinidin as main pigment, contains flavone C-glycoside	Average: 320.19°, bluest point: 294.53°

SEQUENCE LISTING

SEQUENCE LISTING

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<213> ORGANISM: *Campanula* medium

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				20					25				30	Pro
Ser	His	His	His	His	Leu	Pro	Pro	Gly	Pro	Thr	Gly	Trp	Pro	Ile
				35				40				45		Ile
Gly	Ala	Leu	Pro	Leu	Leu	Gly	Thr	Met	Pro	His	Val	Ser	Leu	Ala
				50			55							60

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Gly	Thr	Val	Val	Ala	Ser	Asn	Pro	Lys	Ala	Ala	Arg	Ala	Phe	Leu	Lys	
				85					90					95		
Thr	His	Asp	Ala	Asn	Phe	Ser	Asn	Arg	Pro	Ile	Asp	Gly	Gly	Pro	Thr	
			100					105					110			
Tyr	Leu	Ala	Tyr	Asn	Ala	Gln	Asp	Met	Val	Phe	Ala	Glu	Tyr	Gly	Pro	
		115					120					125				
Lys	Trp	Lys	Leu	Leu	Arg	Lys	Leu	Cys	Ser	Leu	His	Met	Leu	Gly	Pro	
		130				135					140					
Lys	Ala	Leu	Glu	Asp	Trp	Ala	His	Val	Lys	Val	Ser	Glu	Val	Gly	His	
145					150					155					160	
Met	Leu	Lys	Glu	Met	Tyr	Glu	Gln	Ser	Ser	Lys	Ser	Val	Pro	Val	Pro	
				165					170					175		
Val	Val	Val	Pro	Glu	Met	Leu	Thr	Tyr	Ala	Met	Ala	Asn	Met	Ile	Gly	
			180					185					190			
Arg	Ile	Ile	Leu	Ser	Arg	Arg	Pro	Phe	Val	Ile	Thr	Ser	Lys	Leu	Asp	
		195					200					205				
Ser	Ser	Ala	Ser	Ala	Ser	Ala	Ser	Val	Ser	Glu	Phe	Gln	Tyr	Met	Val	
		210				215					220					
Met	Glu	Leu	Met	Arg	Met	Ala	Gly	Leu	Phe	Asn	Ile	Gly	Asp	Phe	Ile	
225					230				235						240	
Pro	Tyr	Ile	Ala	Trp	Met	Asp	Leu	Gln	Gly	Ile	Gln	Arg	Asp	Met	Lys	
			245						250					255		
Val	Ile	Gln	Lys	Lys	Phe	Asp	Val	Leu	Leu	Asn	Lys	Met	Ile	Lys	Glu	
			260					265					270			
His	Thr	Glu	Ser	Ala	His	Asp	Arg	Lys	Asp	Asn	Pro	Asp	Phe	Leu	Asp	
		275					280					285				
Ile	Leu	Met	Ala	Ala	Thr	Gln	Glu	Asn	Thr	Glu	Gly	Ile	Gln	Leu	Asn	
		290				295					300					
Leu	Val	Asn	Val	Lys	Ala	Leu	Leu	Leu	Asp	Leu	Phe	Thr	Ala	Gly	Thr	
305					310					315					320	
Asp	Thr	Ser	Ser	Ser	Val	Ile	Glu	Trp	Ala	Leu	Ala	Glu	Met	Leu	Asn	
			325						330					335		
His	Arg	Gln	Ile	Leu	Asn	Arg	Ala	His	Glu	Glu	Met	Asp	Gln	Val	Ile	
			340				345						350			
Gly	Arg	Asn	Arg	Arg	Leu	Glu	Gln	Ser	Asp	Ile	Pro	Asn	Leu	Pro	Tyr	
		355				360						365				
Phe	Gln	Ala	Ile	Cys	Lys	Glu	Thr	Phe	Arg	Lys	His	Pro	Ser	Thr	Pro	
		370				375				380						
Leu	Asn	Leu	Pro	Arg	Ile	Ser	Thr	Glu	Ala	Cys	Glu	Val	Asp	Gly	Phe	
385					390				395						400	
His	Ile	Pro	Lys	Asn	Thr	Arg	Leu	Ile	Val	Asn	Ile	Trp	Ala	Ile	Gly	
				405				410						415		
Arg	Asp	Pro	Lys	Val	Trp	Glu	Asn	Pro	Leu	Asp	Phe	Thr	Pro	Glu	Arg	
			420					425					430			
Phe	Leu	Ser	Glu	Lys	His	Ala	Lys	Ile	Asp	Pro	Arg	Gly	Asn	His	Phe	
		435					440					445				
Glu	Leu	Ile	Pro	Phe	Gly	Ala	Gly	Arg	Arg	Ile	Cys	Ala	Gly	Ala	Arg	
		450				455					460					
Met	Gly	Ala	Ala	Ser	Val	Glu	Tyr	Ile	Leu	Gly	Thr	Leu	Val	His	Ser	
465					470					475					480	

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Phe Asp Trp Lys Leu Pro Asp Gly Val Val Glu Val Asn Met Glu Glu
485 490 495

Ser Phe Gly Ile Ala Leu Gln Lys Lys Val Pro Leu Ser Ala Ile Val
500 505 510

Thr Pro Arg Leu Pro Pro Ser Ser Tyr Thr Val
515 520

<210> SEQ ID NO 3
<211> LENGTH: 720
<212> TYPE: DNA
<213> ORGANISM: Torenia hybrid cultivar

<400> SEQUENCE: 3

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atgaaagata agttctatgg caccattttg cagagcgaag ccctcgcaaa gtatctgtta      60
gagacaagtg cctatccacg agaacatccg cagctcaaag aactaaggag cgcaactgtg      120
gacaagtatc aatattggag cttgatgaat gttccagctg atgaggggca gttcatttca      180
atgttactga aaattatgaa cgcaaaaaag acaattgaag ttggagtttt cacaggctac      240
tcactcctat caactgctct ggctctacct gatgatggca aaatcgttgc cattgatcct      300
gatagagaag cttatgagac tggtttgcca tttatcaaga aagcaaacgt ggctcataaa      360
atccaataca tacaatctga tgccatgaaa gtcattgaatg acctcattgc tgccaaggga      420
gaagaagaag aggggagctt tgactttggg ttcgtggatg cagacaaaga aaactacata      480
aactaccacg agaaactggt gaagctgggt aaggttgagg ggatcatagg atacgacaac      540
actctgtggt ctggaacagt tgctgcatct gaagacgatg agaataatat gcaagactac      600
ttaagagggt gcagagggca taccctcaaa ctaaactcct ttctcgcaaa cgatgatcgg      660
attgaattgg ctcacctctc tattggagat ggactcacct tgtgcaaacg tctcaaataa      720

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<210> SEQ ID NO 4
<211> LENGTH: 239
<212> TYPE: PRT
<213> ORGANISM: Torenia hybrid cultivar

<400> SEQUENCE: 4

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Met Lys Asp Lys Phe Tyr Gly Thr Ile Leu Gln Ser Glu Ala Leu Ala      15
1      5      10      15
Lys Tyr Leu Leu Glu Thr Ser Ala Tyr Pro Arg Glu His Pro Gln Leu      30
20      25      30
Lys Glu Leu Arg Ser Ala Thr Val Asp Lys Tyr Gln Tyr Trp Ser Leu      45
35      40      45
Met Asn Val Pro Ala Asp Glu Gly Gln Phe Ile Ser Met Leu Leu Lys      60
50      55      60
Ile Met Asn Ala Lys Lys Thr Ile Glu Val Gly Val Phe Thr Gly Tyr      80
65      70      75      80
Ser Leu Leu Ser Thr Ala Leu Ala Leu Pro Asp Asp Gly Lys Ile Val      95
85      90      95
Ala Ile Asp Pro Asp Arg Glu Ala Tyr Glu Thr Gly Leu Pro Phe Ile      110
100      105      110
Lys Lys Ala Asn Val Ala His Lys Ile Gln Tyr Ile Gln Ser Asp Ala      125
115      120      125
Met Lys Val Met Asn Asp Leu Ile Ala Ala Lys Gly Glu Glu Glu Glu      140
130      135      140
Gly Ser Phe Asp Phe Gly Phe Val Asp Ala Asp Lys Glu Asn Tyr Ile      160
145      150      155      160

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Asn	Tyr	His	Glu	Lys	Leu	Leu	Lys	Leu	Val	Lys	Val	Gly	Gly	Ile	Ile
				165					170					175	
Gly	Tyr	Asp	Asn	Thr	Leu	Trp	Ser	Gly	Thr	Val	Ala	Ala	Ser	Glu	Asp
			180					185					190		
Asp	Glu	Asn	Asn	Met	Gln	Asp	Tyr	Leu	Arg	Gly	Cys	Arg	Gly	His	Ile
		195					200					205			
Leu	Lys	Leu	Asn	Ser	Phe	Leu	Ala	Asn	Asp	Asp	Arg	Ile	Glu	Leu	Ala
	210					215					220				
His	Leu	Ser	Ile	Gly	Asp	Gly	Leu	Thr	Leu	Cys	Lys	Arg	Leu	Lys	
225					230					235					

<210> SEQ ID NO 5
 <211> LENGTH: 1920
 <212> TYPE: DNA
 <213> ORGANISM: Glycyrrhiza echinata

<400> SEQUENCE: 5

ccccaat	ctctatcata	agccattccg	ttgattgagc	ttcctttccg	tgaaaaaat	60
aactaagcga	tatggaacct	caactcgtag	cagtgtctgt	gttggtttca	gcacttatct	120
gctacttctt	tttcoggcca	tatttccacc	gttacggaaa	aaaccttcca	ccatctcctt	180
ttttccggct	tccaataatt	ggccacatgc	acatgttagg	ttcccttctt	caccaatcct	240
tccacaacct	ctcccacogt	tacggtctct	tgttttcact	taactttggc	tctgttctct	300
gtgtcgttgc	ttcaaccctt	cactttgcca	aacaactcct	tcaaaccaac	gaactcgcct	360
ttaactgtcg	cattgaatca	accgccgtta	aaaaactcac	ttacgagtct	tccttggcct	420
tgcacacctta	cgggtattac	tggaggttca	ttaagaagct	gagcatgaac	gagcttttgg	480
gctctcgtag	cataaacaac	ttccaacacc	tacgagcaca	agagacccat	caattgttaa	540
ggcttttgtc	caacagggca	agagcgtttg	aggccgtgaa	tatcaccgag	gagcttctta	600
agttgaccaa	caacgttatt	tctataatga	tgggtgggga	ggcagaggag	gcaagggatg	660
tgggtgcgtga	tgtgacggag	atatttgag	agttaaatgt	ttcggatttt	atttggttgt	720
ttaagaagat	ggacttgcag	gggtttggga	agaggattga	ggatttgttt	cagaggtttg	780
atacgttggt	ggaaaggatt	attagcaagc	gggagcagac	gaggaaagac	agaaggagga	840
atgggaagaa	gggtgagcag	gggagtgggt	atgggatcag	agactttctt	gatattctgc	900
ttgactgtac	tgaggatgag	aattccgaga	ttaaaatcca	aagggttcac	attaaggcct	960
tgattatgga	tttcttcact	gcagggacgg	ataccacagc	gatttcaaca	gagtgggcat	1020
tagtggagct	cgtcaagaaa	ccctccgtgc	tacaaaaagt	tcgtgaagag	atagacaatg	1080
tcgtaggaaa	agacagactt	gttgaggaat	ctgattgtcc	taatctccca	tatctccaag	1140
ccattcttaa	agaaacattc	cgtttgcacc	caccggttcc	tatgggttaca	agaagatgcg	1200
tggcagagtg	cacggtagag	aattacgtca	tcccagaaga	ctcacttctc	tttgtgaatg	1260
tttgggccat	cggagaagaa	ccaaagtgtt	gggacaaccc	attggagtgt	cgccccgaac	1320
gattcttaaa	actagaagga	gattccagtg	gagttgttga	tgtgagggga	agccattttc	1380
agcttctgcc	atttgggtct	ggaaggagga	tgtgccctgg	tgtgtccttg	gctatgcaag	1440
aggtgccagc	actacttggt	gctataatcc	agtgctttga	tttccacgtt	gtgggtccca	1500
aaggtgagat	tttgaagggt	gatgacatag	tcattaatgt	ggatgaaagg	ccaggattga	1560
cggctccaag	ggcacataac	cttggtgtgt	ttcccgttga	tagaacaagt	ggcgggtggac	1620
ccctcaaaat	cattgaatgt	tgattattcg	tctcttgaat	ttgatctgt	gtgtggcttt	1680

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gaataacatg tatgggtgat gtagtatgt gttcttttct ttctttctaa ttctgatcag 1740
tagcgtagac taggcactag ccttcgtag taggcaataac ttccggcaaat taacgaacat 1800
atgctgttca gagatatttt ttgccatgta tcgtcattct attctaggtt gtttttccgt 1860
tttccttatt acattctatg ataaatataa taaattgagt aatattatag tctcttaatt 1920

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<210> SEQ ID NO 6
<211> LENGTH: 523
<212> TYPE: PRT
<213> ORGANISM: Glycyrrhiza echinata

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<400> SEQUENCE: 6

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Met Glu Pro Gln Leu Val Ala Val Ser Val Leu Val Ser Ala Leu Ile
 1             5             10             15
Cys Tyr Phe Phe Phe Arg Pro Tyr Phe His Arg Tyr Gly Lys Asn Leu
          20             25             30
Pro Pro Ser Pro Phe Phe Arg Leu Pro Ile Ile Gly His Met His Met
          35             40             45
Leu Gly Pro Leu Leu His Gln Ser Phe His Asn Leu Ser His Arg Tyr
 50             55             60
Gly Pro Leu Phe Ser Leu Asn Phe Gly Ser Val Leu Cys Val Val Ala
 65             70             75             80
Ser Thr Pro His Phe Ala Lys Gln Leu Leu Gln Thr Asn Glu Leu Ala
          85             90             95
Phe Asn Cys Arg Ile Glu Ser Thr Ala Val Lys Lys Leu Thr Tyr Glu
          100            105            110
Ser Ser Leu Ala Phe Ala Pro Tyr Gly Asp Tyr Trp Arg Phe Ile Lys
          115            120            125
Lys Leu Ser Met Asn Glu Leu Leu Gly Ser Arg Ser Ile Asn Asn Phe
          130            135            140
Gln His Leu Arg Ala Gln Glu Thr His Gln Leu Leu Arg Leu Leu Ser
          145            150            155            160
Asn Arg Ala Arg Ala Phe Glu Ala Val Asn Ile Thr Glu Glu Leu Leu
          165            170            175
Lys Leu Thr Asn Asn Val Ile Ser Ile Met Met Val Gly Glu Ala Glu
          180            185            190
Glu Ala Arg Asp Val Val Arg Asp Val Thr Glu Ile Phe Gly Glu Phe
          195            200            205
Asn Val Ser Asp Phe Ile Trp Leu Phe Lys Lys Met Asp Leu Gln Gly
          210            215            220
Phe Gly Lys Arg Ile Glu Asp Leu Phe Gln Arg Phe Asp Thr Leu Val
          225            230            235            240
Glu Arg Ile Ile Ser Lys Arg Glu Gln Thr Arg Lys Asp Arg Arg Arg
          245            250            255
Asn Gly Lys Lys Gly Glu Gln Gly Ser Gly Asp Gly Ile Arg Asp Phe
          260            265            270
Leu Asp Ile Leu Leu Asp Cys Thr Glu Asp Glu Asn Ser Glu Ile Lys
          275            280            285
Ile Gln Arg Val His Ile Lys Ala Leu Ile Met Asp Phe Phe Thr Ala
          290            295            300
Gly Thr Asp Thr Thr Ala Ile Ser Thr Glu Trp Ala Leu Val Glu Leu
          305            310            315            320
Val Lys Lys Pro Ser Val Leu Gln Lys Val Arg Glu Glu Ile Asp Asn
          325            330            335

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Val	Val	Gly	Lys	Asp	Arg	Leu	Val	Glu	Glu	Ser	Asp	Cys	Pro	Asn	Leu
			340					345					350		
Pro	Tyr	Leu	Gln	Ala	Ile	Leu	Lys	Glu	Thr	Phe	Arg	Leu	His	Pro	Pro
		355					360					365			
Val	Pro	Met	Val	Thr	Arg	Arg	Cys	Val	Ala	Glu	Cys	Thr	Val	Glu	Asn
		370				375					380				
Tyr	Val	Ile	Pro	Glu	Asp	Ser	Leu	Leu	Phe	Val	Asn	Val	Trp	Ser	Ile
385					390					395					400
Gly	Arg	Asn	Pro	Lys	Phe	Trp	Asp	Asn	Pro	Leu	Glu	Phe	Arg	Pro	Glu
				405					410					415	
Arg	Phe	Leu	Lys	Leu	Glu	Gly	Asp	Ser	Ser	Gly	Val	Val	Asp	Val	Arg
			420				425						430		
Gly	Ser	His	Phe	Gln	Leu	Leu	Pro	Phe	Gly	Ser	Gly	Arg	Arg	Met	Cys
		435					440					445			
Pro	Gly	Val	Ser	Leu	Ala	Met	Gln	Glu	Val	Pro	Ala	Leu	Leu	Gly	Ala
		450				455					460				
Ile	Ile	Gln	Cys	Phe	Asp	Phe	His	Val	Val	Gly	Pro	Lys	Gly	Glu	Ile
465					470					475					480
Leu	Lys	Gly	Asp	Asp	Ile	Val	Ile	Asn	Val	Asp	Glu	Arg	Pro	Gly	Leu
			485					490						495	
Thr	Ala	Pro	Arg	Ala	His	Asn	Leu	Val	Cys	Val	Pro	Val	Asp	Arg	Thr
			500				505						510		
Ser	Gly	Gly	Gly	Pro	Leu	Lys	Ile	Ile	Glu	Cys					
		515				520									

<210> SEQ ID NO 7

<211> LENGTH: 1416

<212> TYPE: DNA

<213> ORGANISM: Oryza sativa Japonica Group

<400> SEQUENCE: 7

atgccgagct ctggcgacgc ggccggcagg cgcccgcatg tgggtgctcat cccgagcgcc	60
ggcatggggc acctgtctcc ctccggccgc ctcccgctgg cgtctctctc cgccacggc	120
tgcgacgtct cctctgtcac ggtgtctccc acggtgtcca ccgaggagtc gaagcacctc	180
gacgcgtgtg tcgacgcgtt cccggcggtg cggcggtctg acttcgagct cgcgcgttc	240
gacgcgtcgg agttccccag cgccgacccg ttctctctcc ggttcgaggc catgcggcgg	300
tcggcgccgc tgcctggccc gctctcacc ggcccgccgc catcggcgct cgccacggac	360
atcgcgtgta catcgtctgt cataccctgt gcgaaggagc agggcctccc gtgccacatc	420
ctcttcaccg cctccgccgc gatgtctctc ctctgcgcct acttccccac atacctcgac	480
gccaaacgtg gcgacggcgg cggcgctggc gacgtcgaca tccccggcgt gtaccgcac	540
cccaaggcct ccattccgca ggcgctgcac gatcccaacc acctcttcac ccgccagttc	600
gtcgccaacg gccggagcct cagcagcgcc gccggcatcc tcgtcaaacac gttcgacgcc	660
ttggagccgg agggcgctgc ggccctgcag caggggcaagg tcgcctccgg ctteccgcgg	720
gtgttcgccg tggggccact tctcccgccg agcaaccagg caaaagatcc gcaggcaaac	780
tacatggagt ggctcgacgc gcagccggcg cggtcggtgg tgtacgtgag cttcggcagc	840
cgcaaggcca tctcagggga gcagctcagg gagctcggcg ccgggtggga gaccagcggc	900
cacagggttc tgtgggtggt gaagagcacc gtcgtggaca gggacgacgc cgccgagctc	960
ggcgagctgc tcggcgaggg gttcttgaag cgggtggaga agcgaggcct cgtcaccagg	1020
gcatgggtgg atcaggaaga ggtcctgaag cagcagtcgg tggcgctggt cgtgagccac	1080

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tgcggctgga actcgggtgac ggaggcggcg gcgagcggcg tcccgggtgct ggcgctgccg 1140
aggttcggcg accagcgggt gaactccggc gtggtggcgc gcgccgggct cggcgtgtgg 1200
gcggacacct ggagctggga gggggaagcc ggggtgatcg gcgcagagga gatatcggag 1260
aaggtgaagg cggcgatggc ggacgaggcg ttgcgtagga aggcggcgag cctcgccaag 1320
gccgccgga aggcgcgcgc cggcgggtgga tcgagccacc gttgtctggt cgagttcgcg 1380
cggtgtgcc aagggggaac atgtgcact aattga 1416

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<210> SEQ ID NO 8

<211> LENGTH: 471

<212> TYPE: PRT

<213> ORGANISM: Oryza sativa Japonica Group

<400> SEQUENCE: 8

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Met Pro Ser Ser Gly Asp Ala Ala Gly Arg Arg Pro His Val Val Leu
1          5          10          15
Ile Pro Ser Ala Gly Met Gly His Leu Val Pro Phe Gly Arg Leu Ala
20        25        30
Val Ala Leu Ser Ser Gly His Gly Cys Asp Val Ser Leu Val Thr Val
35        40        45
Leu Pro Thr Val Ser Thr Ala Glu Ser Lys His Leu Asp Ala Leu Phe
50        55        60
Asp Ala Phe Pro Ala Val Arg Arg Leu Asp Phe Glu Leu Ala Pro Phe
65        70        75        80
Asp Ala Ser Glu Phe Pro Ser Ala Asp Pro Phe Phe Leu Arg Phe Glu
85        90        95
Ala Met Arg Arg Ser Ala Pro Leu Leu Gly Pro Leu Leu Thr Gly Ala
100       105       110
Gly Ala Ser Ala Leu Ala Thr Asp Ile Ala Leu Thr Ser Val Val Ile
115       120       125
Pro Val Ala Lys Glu Gln Gly Leu Pro Cys His Ile Leu Phe Thr Ala
130       135       140
Ser Ala Ala Met Leu Ser Leu Cys Ala Tyr Phe Pro Thr Tyr Leu Asp
145       150       155       160
Ala Asn Ala Gly Asp Gly Gly Gly Val Gly Asp Val Asp Ile Pro Gly
165       170       175
Val Tyr Arg Ile Pro Lys Ala Ser Ile Pro Gln Ala Leu His Asp Pro
180       185       190
Asn His Leu Phe Thr Arg Gln Phe Val Ala Asn Gly Arg Ser Leu Thr
195       200       205
Ser Ala Ala Gly Ile Leu Val Asn Thr Phe Asp Ala Leu Glu Pro Glu
210       215       220
Ala Val Ala Ala Leu Gln Gln Gly Lys Val Ala Ser Gly Phe Pro Pro
225       230       235       240
Val Phe Ala Val Gly Pro Leu Leu Pro Ala Ser Asn Gln Ala Lys Asp
245       250       255
Pro Gln Ala Asn Tyr Met Glu Trp Leu Asp Ala Gln Pro Ala Arg Ser
260       265       270
Val Val Tyr Val Ser Phe Gly Ser Arg Lys Ala Ile Ser Gly Glu Gln
275       280       285
Leu Arg Glu Leu Ala Ala Gly Leu Glu Thr Ser Gly His Arg Phe Leu
290       295       300
Trp Val Val Lys Ser Thr Val Val Asp Arg Asp Asp Ala Ala Glu Leu

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305	310	315	320
Gly Glu Leu Leu Gly Glu Gly Phe Leu Lys Arg Val Glu Lys Arg Gly	325	330	335
Leu Val Thr Lys Ala Trp Val Asp Gln Glu Glu Val Leu Lys His Glu	340	345	350
Ser Val Ala Leu Phe Val Ser His Cys Gly Trp Asn Ser Val Thr Glu	355	360	365
Ala Ala Ala Ser Gly Val Pro Val Leu Ala Leu Pro Arg Phe Gly Asp	370	375	380
Gln Arg Val Asn Ser Gly Val Val Ala Arg Ala Gly Leu Gly Val Trp	385	390	395
Ala Asp Thr Trp Ser Trp Glu Gly Glu Ala Gly Val Ile Gly Ala Glu	405	410	415
Glu Ile Ser Glu Lys Val Lys Ala Ala Met Ala Asp Glu Ala Leu Arg	420	425	430
Arg Lys Ala Ala Ser Leu Ala Lys Ala Ala Ala Lys Ala Val Ala Gly	435	440	445
Gly Gly Ser Ser His Arg Cys Leu Val Glu Phe Ala Arg Leu Cys Gln	450	455	460
Gly Gly Thr Cys Arg Thr Asn	465	470	

<210> SEQ ID NO 9
 <211> LENGTH: 990
 <212> TYPE: DNA
 <213> ORGANISM: Lotus japonicus

<400> SEQUENCE: 9

atggtcttctg caacaaccac cccaacaaa gagatagaca gagacctccc tctctttctc	60
cgagtctaca aagatggaac cgtggaacgt ctcttaggct ctctgtcgt tccagcaatc	120
cctcatgacc cagaacacaga ggtctcatca aaagacatag tcatctcaca aaccccccta	180
atctctgctc gtatocacct cccaaaacag agcaaccccc aaaacccaaa ggttccaata	240
ttgatctact accatggtgg tgcgttttgc cttgaatcag ctttctcctt cctccaccaa	300
cgctacctca acatcggtggc ttacgatca aacgttgtgg tggtttcctg cgagtacagg	360
ctcgccagc agcatcctct gcctgcagca tatgaagatg gttgggaagc tctgaaatgg	420
gttacctctc attccaccga caacaaaccc atcaactctg agccatgggt gatcgaacat	480
ggtgatttca gcagattcta catcgagggt gacacttcag gtgccaacat cgcataccat	540
gtgggtctcc gtgtcggtgg tggagttgag aaattgccag gggatgtgaa aattgcaggg	600
gcgttacttg cttttccctt gttttggagt tcataccctg ttttgaaga acctgttgag	660
gggtttgaac agagtttgag caggaagggt tggaaacttg tgtaccaga tgcacctggt	720
gggatcgaca accctctgat caatcctttg gctgatgggg ctccaagctt gacaacgttt	780
ggaagcaaca agatgttgat ctttgttgca gggaatgatg aactgagaga cagaggaatc	840
tggttctatg aggctgtgaa gaagagtgg tgggaagggt atgtggaact cattcgagt	900
gatggagagg agcattgctt ccagatttac catcctgaat ctgagaattc taaagacatg	960
atgaagcgca tagcttcttt ccttgtttga	990

<210> SEQ ID NO 10
 <211> LENGTH: 329
 <212> TYPE: PRT
 <213> ORGANISM: Lotus japonicus

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<400> SEQUENCE: 10

Met Ala Ser Ala Thr Thr Thr Pro Thr Lys Glu Ile Asp Arg Asp Leu
 1 5 10 15
 Pro Pro Leu Leu Arg Val Tyr Lys Asp Gly Thr Val Glu Arg Leu Leu
 20 25 30
 Gly Ser Pro Val Val Pro Ala Ile Pro His Asp Pro Glu Thr Glu Val
 35 40 45
 Ser Ser Lys Asp Ile Val Ile Ser Gln Thr Pro Leu Ile Ser Ala Arg
 50 55 60
 Ile His Leu Pro Lys Gln Ser Asn Pro Gln Asn Pro Lys Val Pro Ile
 65 70 75 80
 Leu Ile Tyr Tyr His Gly Gly Ala Phe Cys Leu Glu Ser Ala Phe Ser
 85 90 95
 Phe Leu His Gln Arg Tyr Leu Asn Ile Val Ala Ser Arg Ser Asn Val
 100 105 110
 Val Val Val Ser Val Glu Tyr Arg Leu Ala Pro Glu His Pro Leu Pro
 115 120 125
 Ala Ala Tyr Glu Asp Gly Trp Glu Ala Leu Lys Trp Val Thr Ser His
 130 135 140
 Ser Thr Asp Asn Lys Pro Ile Asn Ser Glu Pro Trp Leu Ile Glu His
 145 150 155 160
 Gly Asp Phe Ser Arg Phe Tyr Ile Gly Gly Asp Thr Ser Gly Ala Asn
 165 170 175
 Ile Ala Tyr His Val Gly Leu Arg Val Gly Gly Gly Val Glu Lys Leu
 180 185 190
 Pro Gly Asp Val Lys Ile Ala Gly Ala Leu Leu Ala Phe Pro Leu Phe
 195 200 205
 Trp Ser Ser Tyr Pro Val Leu Glu Glu Pro Val Glu Gly Phe Glu Gln
 210 215 220
 Ser Leu Ser Arg Lys Val Trp Asn Phe Val Tyr Pro Asp Ala Pro Gly
 225 230 235 240
 Gly Ile Asp Asn Pro Leu Ile Asn Pro Leu Ala Asp Gly Ala Pro Ser
 245 250 255
 Leu Thr Thr Phe Gly Ser Asn Lys Met Leu Ile Phe Val Ala Gly Asn
 260 265 270
 Asp Glu Leu Arg Asp Arg Gly Ile Trp Phe Tyr Glu Ala Val Lys Lys
 275 280 285
 Ser Glu Trp Glu Gly Asp Val Glu Leu Ile Arg Val Asp Gly Glu Glu
 290 295 300
 His Cys Phe Gln Ile Tyr His Pro Glu Ser Glu Asn Ser Lys Asp Met
 305 310 315 320
 Met Lys Arg Ile Ala Ser Phe Leu Val
 325

<210> SEQ ID NO 11

<211> LENGTH: 1374

<212> TYPE: DNA

<213> ORGANISM: Fagopyrum esculentum

<400> SEQUENCE: 11

atgatgggag acttaaccac atcttttcca gcaaccacac tcaccacgaa tgatcaacct 60
 cacgtggtgg tttgctcagg agctggtatg gggcatttga caccgttcct taacttgccc 120
 tctgctcttt catctgcgcc gtataactgc aagggtactc tggtaatcgt gattccgctt 180

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atcacagatg ccgaaagtca ccacattagc agttttttct cttcacatcc cactatccac   240
aggctagact tccacgtgaa tttaccggct ccaaagccca atgtggatcc gttctttctc   300
cgttacaaga gcattctcaga ctctgcacat agacttcctg tccacttatc ggccttgagt   360
ccccgatata ctgcggtttt ctctgacttc ctcttcactc agggctctgaa cactactttg   420
cctcatcttc ccaactacac gtttactacg acttctgctc gggtttttcac cttgatgtcc   480
tatgttccto acttggtccaa atcgagctca tcaagtccctg tagagattcc tggccttgag   540
cggtttccaa ctgacaacat cctccccctt tttttcaacc cagagcacat ctttaccagt   600
ttcaccatat ccaatgccaa gtacttctcg ttgagcaaag ggattttggt gaataccttc   660
gactcctttg agcccgagac actctctgca ttgaactcgg gagacacatt gtcagatctc   720
cctccggtca tccctattgg cccactgaac gaactcgagc ataacaaaca ggaggaactc   780
ctgccatggc tcgaccaaca acctgagaaa tccgtccttt acgtctcttt tgggaatcga   840
acagccatga gtacgcagca gattctcgag ctaggaatgg gattagagcg tagtgattgt   900
cgcttttatt gggctgttaa gacgtccaag atagacaagg acgataagag tgagctacgt   960
aaactctttg gtgaagagct ctaccttaaa ctgtccgaga agggaaagct tgtgaaatgg  1020
gtgaacaaaa ccgagattct cgggcatact gctgtagggt gtttctctgc tcattgtggg  1080
tggaattcgg tcattggaagc agccagaaga ggtgttccga tattggcttg gccacaacac  1140
ggatgatcagc gggaaaatgc atgggttgtt gaaaaggctg gcttgggagt gtgggaaagg  1200
gagtgggcaa gcgggattca ggcagccatt gttgagaagg tcaagatgat catgggcaat  1260
aacgaccttc gcaaaagcgc tatgaagggt ggtgaagagg caaagcgagc atgtgatgta  1320
ggtgggagtt ccgctaccgc cttgatgaat attatcgggt cattgaagag gtaa      1374

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<210> SEQ ID NO 12

<211> LENGTH: 457

<212> TYPE: PRT

<213> ORGANISM: *Fagopyrum esculentum*

<400> SEQUENCE: 12

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Leu Thr Pro Phe Leu Asn Leu Ala Ser Ala Leu Ser Ser Ala Pro Tyr
 35             40            45

Asn Cys Lys Val Thr Leu Leu Ile Val Ile Pro Leu Ile Thr Asp Ala
 50             55            60

Glu Ser His His Ile Ser Ser Phe Phe Ser Ser His Pro Thr Ile His
 65             70            75            80

Arg Leu Asp Phe His Val Asn Leu Pro Ala Pro Lys Pro Asn Val Asp
 85             90            95

Pro Phe Phe Leu Arg Tyr Lys Ser Ile Ser Asp Ser Ala His Arg Leu
100            105           110

Pro Val His Leu Ser Ala Leu Ser Pro Pro Ile Ser Ala Val Phe Ser
115            120           125

Asp Phe Leu Phe Thr Gln Gly Leu Asn Thr Thr Leu Pro His Leu Pro
130            135           140

Asn Tyr Thr Phe Thr Thr Thr Ser Ala Arg Phe Phe Thr Leu Met Ser
145            150           155           160

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-continued

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<210> SEQ ID NO 15

<211> LENGTH: 58

<212> TYPE: DNA

<213> ORGANISM: Arabidopsis thaliana

<400> SEQUENCE: 15

tacatcacia tcacacaaaa ctaacaaaag atcaaaagca agttcttcac tgttgata	58
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The invention claimed is:

1. A buckwheat-derived C-glucosyltransferase (CGT) gene or its homolog, wherein:

the buckwheat-derived CGT gene or its homolog is a polynucleotide consisting of the nucleotide sequence as set forth in SEQ ID NO: 11.

2. The buckwheat-derived CGT gene or its homolog according to claim 1, wherein the *Arabidopsis thaliana* alcohol dehydrogenase (ADH) gene-derived untranslated region (5'-UTR) (SEQ ID NO: 15) or the *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) has been added.

3. A vector comprising a buckwheat-derived CGT gene or its homolog, wherein the buckwheat-derived CGT gene or its homolog is selected from the group consisting of:

- (a) a polynucleotide consisting of the nucleotide sequence as set forth in SEQ ID NO: 11;
- (b) a polynucleotide that encodes a protein consisting of the amino acid sequence as set forth in SEQ ID NO: 12; and

(c) a polynucleotide that encodes a protein having an amino acid sequence with at least 98% identity with respect to the amino acid sequence as set forth in SEQ ID NO: 12 and having the same activity as a protein encoded by the polynucleotide of (b), and

wherein the vector further comprises a flavanone 2-hydroxylase (F2H) gene or its homolog, and a dehydratase (FDH) gene or its homolog.

4. The vector according to claim 3, wherein the *Arabidopsis thaliana* ADH gene-derived untranslated region (5'-UTR) (SEQ ID NO: 15) or the *Arabidopsis thaliana* HSPRO gene-derived untranslated region (5'-UTR) (SEQ ID NO: 13) has been added to the buckwheat-derived CGT gene or its homolog.

5. The vector according to claim 3, wherein the F2H gene or its homolog is selected from the group consisting of:

- (a) a polynucleotide consisting of the nucleotide sequence as set forth in SEQ ID NO: 5;

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- (b) a polynucleotide that encodes a protein consisting of the amino acid sequence as set forth in SEQ ID NO: 6; and
- (c) a polynucleotide that encodes a protein having an amino acid sequence with at least 98% identity with respect to the amino acid sequence as set forth in SEQ ID NO: 6 and having the same activity as a protein encoded by the polynucleotide of (b), and the FDH gene or its homolog is selected from the group consisting of:
- (a) a polynucleotide consisting of the nucleotide sequence as set forth in SEQ ID NO: 9;
- (b) a polynucleotide that encodes a protein consisting of the amino acid sequence as set forth in SEQ ID NO: 10; and
- (c) a polynucleotide that encodes a protein having an amino acid sequence with at least 98% identity with respect to the amino acid sequence as set forth in SEQ ID NO: 10 and having the same activity as a protein encoded by the polynucleotide of (3b).
6. A transgenic rose plant or its inbred or outbred progeny, a propagule thereof, a partial plant body thereof, a tissue or a cell of the transgenic rose plant thereof, a cut flower of the transgenic rose plant thereof comprising the buckwheat-derived CGT gene or its homolog according to claim 1.
7. The transgenic plant or its inbred or outbred progeny, a propagule thereof, a partial plant body thereof, a tissue or

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a cell of the transgenic rose plant thereof, a cut flower of the transgenic rose plant thereof according to claim 6, wherein the cut flower of the transgenic rose plant is a processed form selected from the group consisting of preserved, dried or resin sealed.

8. A method for creating transgenic rose plants with blue flower color, comprising a step of transferring the buckwheat-derived C-glucosyltransferase (CGT) gene according to claim 1 into a host rose plant to cause delphinidin-type anthocyanins and flavone mono-C-glycosides to coexist in the rose plant cells.

9. The method according to claim 8, wherein the flavone mono-C-glycoside is apigenin 6-C-glucoside, luteolin 6-C-glucoside, tricetin 6-C-glucoside, apigenin 8-C-glucoside, luteolin 8-C-glucoside or tricetin 8-C-glucoside, or a derivative thereof.

10. The method according to claim 8, wherein the delphinidin-type anthocyanin is selected from the group consisting of malvidins, delphinidins, petunidins and their combinations.

11. A method for creating transgenic rose plants with blue flower color, comprising a step of transferring the vector according to claim 3 into a host plant to cause delphinidin-type anthocyanins and flavone mono-C-glycosides to coexist in the plant cells.

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